Borrow Long, Dump Short: Debt Maturity and Leverage Choices under Negative Interest Rates†

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Abstract

We study the impact of negative interest rate policy (NIRP) on corporate debt structure using a difference-in-differences design applied to a panel of over 54,000 firms across 127 countries. We find that firms in NIRP jurisdictions systematically reallocate from short-term to long-term debt—a pattern we term "borrow long, dump short"—consistent with locking in historically low long-term borrowing costs. This maturity rebalancing is most pronounced among smaller, older, highly leveraged, and less liquid firms. Notably, aggregate leverage remains stable on impact but rises significantly when negative rates persist. To rationalise these findings, we develop a DSGE model calibrated to the euro area that identifies two transmission channels: a *signalling channel* through which NIRP compresses real long-term yields, inducing immediate maturity rebalancing; and a *reserve-carry channel* through which the deposit rate floor affects bank net worth and credit rapidly reshape debt maturity composition. Our results demonstrate that NIRP can rapidly reshape the maturity composition of corporate debt, but its effect on aggregate leverage depends on policy persistence and credible forward guidance.

JEL classification: E52; E58; G32; E44

Keywords: monetary policy; negative interest rate policy (NIRP); corporate financing; capital structure

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†We would like to thank Xavier D'Haultfoeuille, Bob Gregory, Bruce Preston, Xiaodong Fan, Ben Wang (discussant), seminar and conference participants at the 39th Annual Australian PhD Conference in Economics and Business (University of New South Wales), and Curtin University for helpful comments and discussions on earlier versions of this paper. Richard is supported by the Curtin Strategic Scholarship. The usual disclaimer applies.

1 Introduction and background

The last two decades have witnessed a profound transformation in the conduct of monetary policy, especially in advanced economies. With policy rates pinned at zero for prolonged periods, central banks have been forced to rely on unconventional tools — quantitative easing (QE), forward guidance (FG), and, in a subset of jurisdictions, negative interest rate policy (NIRP) — to provide additional accommodation when the effective lower bound rendered conventional rate cuts infeasible. While the macroeconomic effects of these policies have been studied, their microeconomic transmission — particularly the transmission of negative interest rates to the financing decisions of non-financial corporations — remains far less understood. This gap is striking because corporate balance sheets constitute a critical link in the monetary transmission mechanism: the cost and maturity structure of debt directly shape investment, employment, and ultimately aggregate demand. Traditionally, central banks pay interest on deposits or reserves they hold for commercial banks. Nevertheless, when rates reached the zero-lower bound (ZLB), some central banks in advanced economies began to charge interest on these deposits (i.e. *excess reserves*) to encourage commercial banks to increase their credit supply and influence the country's interest rates, a concept known as negative interest rate policy. Thus, negative policy rates, by pushing a large segment of the yield curve into uncharted territory, create novel incentives and frictions for firms and banks alike, making it essential to examine how corporations adjust their liability side in response.

Despite a rich literature on firms' capital structure (*i.e.*, choosing between debt, equity or some hybrid form of securities) dating back to Modigliani and Miller (1958), surprisingly little is known about how unconventional monetary policy — in particular NIRP — affects the maturity composition of corporate debt. Existing work has largely focused on the aggregate leverage response or on the bank-lending channel of NIRP (Bottero et al., 2022; Demiralp et al., 2021), but the possibility that negative rates trigger a systematic debt-maturity restructuring — firms actively "borrowing long and dumping short" to lock in low long-term rates while avoiding repeated rollover risk — has received almost no attention. This is a critical omission. Debt maturity is not neutral: short-term debt exposes firms to rollover risk and amplifies financial accelerator mechanisms, whereas long-term debt insulates real activity from monetary tightening and liquidity shocks. If NIRP induces a broad-based shift from short-term to long-term liabilities, it could significantly enhance corporate resilience and alter the potency of future monetary policy; yet, the empirical magnitude, causal identification, and underlying mechanisms of such a maturity rebalancing remain unexplored.

This paper fills these gaps by exploiting the European Central Bank's (ECB) adoption of NIRP in June 2014 in a difference-in-differences framework to test the debt maturity and leverage

choices of non-financial firms. Due to prolonged low economic activity, the ECB adopted NIRP in addition to other unconventional monetary policy tools. The ECB reduced its official policy rate (*i.e.*, deposit facility rate, DFR) to zero from 2012 through 2013 (see Figure 1).

The ECB added NIRP to the basket of its unconventional monetary policy tools with the core aim of achieving policy easing—*i.e.*, to stimulate economic activities in the euro area following sustained periods of low aggregate demand and undesirable low inflation. This policy was implemented in 2014 when the ECB reduced the deposit facility rate below 0% to -0.1% (Figure 1). From Figure 1, we can see that before 2014, the deposit facility rate remained at the ZLB. In 2014, when the NIRP was implemented, the rates fell below zero and remained so until 2022. During this period, the corporate loan rate followed a very similar pattern. This has direct cost implications for banks, especially those that hold excess reserves with their national central banks (NCBS).[1] Indeed, Claeys (2021) observes that these costs have been increasing, especially after the introduction of the pandemic emergency purchase program (PEPP) in 2020.

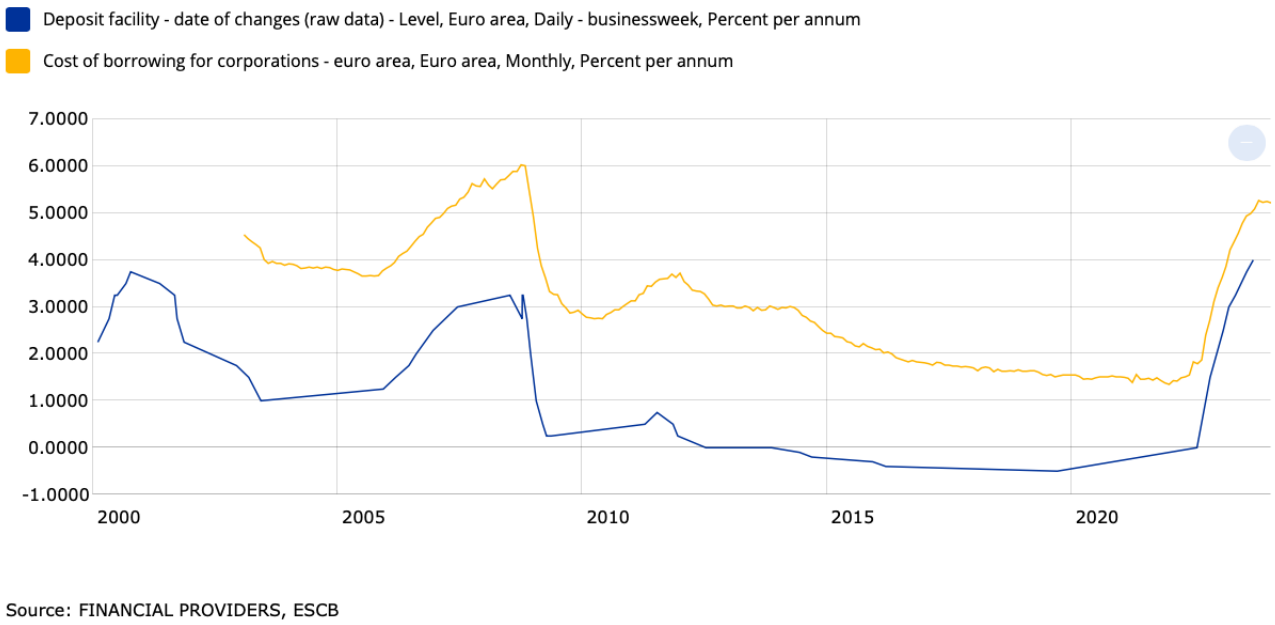

Figure 1: ECB deposit facility rate (%): level and cost of borrowing (%) for corporations

The implication and effectiveness of this policy are still unclear and at an infant stage of empirical enquiry (Bottero et al., 2022; Eggertsson et al., 2024). We identify two key scenarios on how the transmission of this new monetary policy tool can be effective. The first scenario is whether banks will pass on this policy through lending or deposit rates (Brunnermeier and Koby, 2018). A pass-through to deposit rates would mean banks would have to charge negative

[1]Excess reserves are those funds above the minimum required reserves (MRR) held at the NCBs set by the ECB.

interest rates on the retail deposits. This is more rigid, and banks are reluctant to use this channel (Heider et al., 2019) due to competition, depositors' preference for cash, or potential legal limitations (Demiralp et al., 2021; Scheiber et al., 2016). Albertazzi et al. (2020) show that a few banks in some euro area countries did actually implement negative interest rates on retail deposits, even though the majority of non-financial corporate retail deposits (about two-thirds) are still stuck at the ZLB, while negative rates on household deposits remained negligible. Indeed, some studies have found that the retail deposit channel could have a contractionary effect on loan supply (Eggertsson et al., 2024; Heider et al., 2019).

An alternative channel is through bank lending, with the hope that banks benefit from higher loan volumes. This means that banks in the euro area are likely to lend more at lower interest rates. As we observe in Figure 1, the period of the 0% deposit facility rate saw corporate rates around 3% until they began a downward trajectory when the negative interest rate was implemented. Bottero et al. (2022) and Demiralp et al. (2021) show that NIRP is expansionary in that, comparatively, banks in the euro area give out more loans in periods of NIRP. As mentioned earlier, instead of focusing on the policy's effectiveness from the supply side—a perspective that has received some, albeit limited, attention in the literature—we aim to understand its impact from the demand side, which has been notably underexplored.

Hence, this investigation addresses a pivotal policy question: What is the impact of negative interest rate policy on corporate financing decisions? This question underpins the motivation of our study, guiding us to adopt a second scenario focused on comprehending the effectiveness of NIRP through the lens of corporate demand for credit. Accordingly, we explore critical questions such as whether firms increase their leverage in a negative interest rate environment and, if so, the magnitude of this increase, alongside the nature of the debt (long-term *versus* short-term) they are inclined to increase.

While studies such as those by Bottero et al. (2022) and Demiralp et al. (2021) show the expansionary nature of NIRP from the supply side, specifically in terms of bank credit growth—the credit/bank lending channel—our analysis shifts the perspective to the balance sheet channel, which we see as more plausible. As indicated by Bernanke and Gertler (1995), the balance sheet channel relates to the impact of monetary policy on firms' balance sheets and income statements. The authors suggest that this channel is more plausible for monetary policy transmission, even though the bank lending channel has its own merits. By adopting this approach, we can gain nuanced insights into the determinants of corporate capital structure decisions within the context of the NIRP. There will likely be significant firm-specific heterogeneity in these decisions: for instance, it raises questions about whether, in negative interest rate environments, larger firms are more predisposed to increase their debt levels compared to smaller firms; whether older firms are more inclined to leverage additional debt than their younger counterparts; and whether firms

with substantial ex-ante liquid assets are more likely to take on further debt. Given the nascent understanding of NIRP's implications for corporate financing decisions, these questions are of paramount importance and remain relatively unexplored.

This paper makes two principal contributions to the literature on unconventional monetary policy and corporate capital structure. First, we provide new causal evidence on the effects of negative interest rate policy on corporate financing decisions. Using a panel of approximately 54,000 firms across 127 countries—comprising over 400,000 firm-year observations—we implement a difference-in-differences design that exploits the ECB's June 2014 policy shift as a quasi-natural experiment. Despite nearly a decade of experience with negative rates, the firm-level implications of NIRP remain surprisingly underexplored. Our findings document a systematic reallocation from short-term to long-term debt among firms in NIRP jurisdictions, a pattern we characterise as "borrow long, dump short." Importantly, this maturity rebalancing occurs immediately upon policy adoption, while overall leverage remains stable in the short run and rises significantly only when the policy persists. These findings establish debt maturity restructuring as a quantitatively important transmission channel of NIRP—one that operates primarily on the demand side of credit markets and has been largely overlooked in the existing literature.

Second, we develop a medium-scale dynamic stochastic general equilibrium (DSGE) model calibrated to euro area data that rationalises these empirical patterns. The model incorporates financial intermediaries subject to leverage constraints, endogenous corporate debt maturity choice, and the key institutional feature distinguishing NIRP from conventional policy: the zero lower bound on retail deposit rates. This deposit floor creates a wedge between policy rates and funding costs, generating a reserve-carry tax on bank balance sheets that governs credit supply dynamics. The model identifies two transmission channels. A *signalling channel* compresses real long-term yields through forward-looking bond pricing, reducing the relative cost of long-maturity financing and inducing immediate maturity rebalancing. A *reserve-carry channel* links bank net worth to credit supply, explaining why total leverage adjusts only gradually and requires policy persistence to expand. The model's impulse responses closely match the empirical patterns from the DiD implementation.

Together, these contributions provide a unified framework for understanding how NIRP transmits to corporate balance sheets. By integrating rigorous causal identification with structural general equilibrium modelling, we bridge the gap between micro-level firm behaviour and macro-financial transmission mechanisms. Our findings carry direct implications for monetary policy design: negative rates can effectively stimulate maturity rebalancing, but sustained implementation with credible forward guidance is necessary to generate aggregate leverage expansion. As central banks assess the legacy of unconventional policies and contemplate future deployments, understanding these transmission channels—and their distinct time horizons—is essential for

both policy calibration and financial stability assessment.

The remainder of the paper is organised as follows. In Section 2, we review some related studies and discuss our contributions in detail. Section 3 shows a description of the data used in the study. In Section 4, we discuss our identification strategy and present the specification of our empirical model. Section 5 presents the empirical results. Section 6 presents our DSGE model calibrated to the euro area to explain firms' debt maturity choices in a negative interest rate environment, while Section 7 concludes and outlines the study's policy implications.

2 Theoretical considerations and review of related studies

Our inquiry is motivated by three foundational theories in the capital structure literature. Dating back to 1958, *trade-off theory*, as posited by Modigliani and Miller (1958) and later refined by Modigliani and Miller (1963), suggests that firms are incentivised to increase debt to capitalise on the tax shield provided by interest deductions, while also mitigating free cash flow issues (Fama and French, 2002). Miller (1977) argues that the detriments of bankruptcy and agency costs, which could potentially negate this tax advantage, are, in fact, minimal. On the contrary, Myers (2001) underscores that firms, in identifying their optimal capital structure, strive to balance exploiting tax benefits and avoiding financial distress. Empirical support for firms augmenting their leverage is provided by Leland (1994), suggesting that more profitable firms are more likely to have higher debt levels. Hence, within our framework, we can test this theory to determine whether more profitable firms are likely to increase their leverage in a negative interest rate environment.

Subsequently, the *pecking order theory* (POT) (Myers and Majluf 1984; Myers 1984; Titman and Wessels 1988) delineates a financing hierarchy in which firms prioritise internal funding over external debt and equity due to the higher costs of issuing new securities. This theory implies that firms' capital structure choices are dictated not by a trade-off of debt's costs and benefits but by available net cash flows (Fama and French, 2002), predisposing more profitable firms to eschew additional debt in favour of internal funds. Moreover, larger firms, with greater access to external funding, are posited to exhibit a higher propensity to take on debt than their smaller counterparts. We provide sub-sample analyses of the relationship between NIRP and capital structure by firm size.

The concept of *market-timing*[2] has also garnered renewed interest, particularly in the context of how firms adjust their capital structure in response to market conditions (Baker and Wurgler, 2002; Frank and Goyal, 2009). This includes *debt market-timing*, where firms are theorised

[2]See, Taggart (1977),Marsh (1979) Marsh (1982) and Myers (1984) for initial discussions and introduction to market-timing theory.

to increase leverage during periods of low interest rates and *vice versa* (Graham and Harvey, 2001; Korajczyk and Levy, 2003; Bolton et al., 2013; Karpavičius and Yu, 2017). The advent of NIRPs presents an unprecedented context for evaluating these dynamics. We propose preliminary hypotheses suggesting that firms have a propensity to increase leverage during periods of negative interest rates, predicated on prolonged NIRP expectations in the euro area, as noted by Claeys (2021). This aligns with the view of Frank and Goyal (2009), who indicate that firms are likely to raise more debt if the market is unusually favourable, even if they do not need the funds.

The literature exploring the interplay among interest rates, business cycles, and capital structure has bifurcated into two primary streams. One perspective considers the cyclical sensitivity of capital structure, with some studies indicating pro-cyclical leverage adjustments (Gertler and Gilchrist, 1993; Baker and Wurgler, 2002; Frank and Goyal, 2004; Jermann and Quadrini, 2006; Cook and Tang, 2010; Gan et al., 2021), while others suggest a counter-cyclical pattern (Korajczyk and Levy, 2003; Halling et al., 2016). For instance, Jermann and Quadrini (2006) show that the financial structure of firms tends to be volatile in response to business cycles. Here, firms, especially those with high debt exposure, tend to restructure their financial exposure during or after recessions. Specifically, firms reduce their capital structure during recessions and increase it during expansions. This aligns with earlier findings by Gertler and Gilchrist (1993), who show that firms, especially large firms, increase their debt following recessions while small firms do not change their capital structure. More recently, Gan et al. (2021) show that in favourable macroeconomic conditions, firms adjust their capital structure faster than in poor macroeconomic conditions.

On the other hand, Halling et al. (2016) examined the relationship between firms' capital structure and the business cycle and found that, in contrast to the literature, firms change their capital structure in a counter-cyclical manner: that is, firms have higher leverage during recessions than expansions. This seems to be inconsistent with the *trade-off* theory, which suggests that in expansions when there is an expectation of low bankruptcy costs, higher taxable income to provide a shield, and more free cash flow (Korajczyk and Levy, 2003), firms would increase their debt. The counter-cyclical nature of capital structure is also confirmed by Korajczyk and Levy (2003), but it was mostly observed for financially unconstrained firms, whereas constrained firms exhibit a pro-cyclical leverage pattern. This means that financially unconstrained firms tend to time their leverage issuance to coincide with periods of good macroeconomic conditions, while constrained firms do not. The authors define financially constrained firms as those that lack sufficient cash to capitalise on investment opportunities or are hindered by severe agency problems when seeking to access financial markets. However, they show that some firm-specific variables tend to predict firms' target leverage, consistent with both the *trade-off* and *pecking order* theories. Specifically, their results show that higher operating income is negatively associated

with target leverage, consistent with the *pecking order* theory.

The other perspective in the literature involves testing the relationship between the interest rate or cost of debt and firms' capital structure. Barry et al. (2008) using debt issued by U.S. corporations, examined how historical interest rates play a role in the relationship between current interest rates and corporate debt. They found that firms issue more debt when current interest rates are relatively lower than historical interest rates. However, in a later study, Karpavičius and Yu (2017) also tested whether U.S. industrial firms alter their capital structure decisions in response to borrowing costs. The authors found that firms typically do not adjust their capital structure in response to interest rates, except during expected recessions. They attribute this weak relationship to the high leverage adjustment costs.

This dichotomy in examining firms' capital structure decisions based on the interest rate or macroeconomic conditions highlights the complexity of financial decision-making in varying macroeconomic climates, underscored by the *trade-off*, *pecking order* and *debt market-timing* theories' predictions regarding firm behaviour under different conditions. For instance, regarding leverage financing and macroeconomic conditions, there is no consensus in the literature on determining a "good" or "bad" market condition. Frank and Goyal (2009), for example, use expected inflation and the term spread as proxies for debt market conditions, and profit after tax growth and GDP growth as "macroeconomic conditions". Korajczyk and Levy (2003) conversely use the growth rate of aggregate profits of non-financial firms, the two-year market return of stock traded on NYSE, AMEX, & NASDAQ, and the rate of the three-month commercial paper over the three-month Treasury bill, to capture macroeconomic conditions. Graham et al. (2015) studied the long history of capital structure of U.S. corporations and indicated four key macroeconomic themes that influence firms' capital structure. These include changes in tax policy, economic uncertainty, developments in the financial sector, managerial incentives, and government borrowing. The authors found that government debt is a significant determinant of firms' capital structure, alongside other macroeconomic variables, including inflation, GDP growth, and the corporate bond yield spread.

These different macroeconomic measures and perspectives may not capture central banks' monetary policy stance. Our study differs from the previous literature in that, by examining the effect of NIRP on capital structure, we can capture both the macroeconomic environment and interest rates. While we have some understanding of the relationship between conventional interest rate adjustments and capital structure, the specific impact of NIRP remains less understood, particularly in the context of potential operational differences from traditional rate cuts (Demiralp et al., 2021). This study aims to bridge this gap by employing a comprehensive cross-country dataset to elucidate the ramifications of NIRP for firms' capital structure decisions, thereby offering a novel perspective on the influence of monetary policy on corporate finance.

3 Data

Following previous studies (Bekaert et al., 2013), we obtain data for nearly 80,000 firms from the Bureau Van Dijk (BVD) OSIRIS database spanning the years 2001 to 2021. OSIRIS provides a comprehensive database covering most publicly listed and major non-listed companies across 127 countries (Dai, 2012). This helps the study provide a broader perspective on firms' capital structure decisions worldwide. As noted by (Dai, 2012), a major advantage of using OSIRIS is that it covers both developed and developing countries, including many small firms. As we posit earlier, firms, particularly larger firms, that have access to the equity market would have a broader choice of capital structure than their counterparts. Hence, we can delineate differences in capital structure decisions between listed and unlisted firms, between small and large firms, and among other firm-specific heterogeneities that may influence their capital structure decisions.

Initial data pre-processing involved, following the usual convention (for example, Adelino et al. 2023), excluding all financial firms (firms with two-digit Standard Industry Classification [SIC] codes from 60 to 69), utility firms (with primary SIC code 49) and public administration firms (90 and 99) and firms with no classification. Financial, utility, and public administration firms are typically regulated, and hence their leverage choices should differ significantly from those of other industrial firms. For this reason, and in line with the literature, we also exclude these firms from our data. We exclude firm-years with a negative book value of equity or missing data for any of our capital structure measures. Again, following Husted et al. (2020) and Adelino et al. (2023), in order to minimise the potential impact of outliers, the data are winsorised at the 1st and 99th percentiles. We are left with around 54,000 firms across 127 countries after the data-cleaning process. We proceed to discuss the data used in the study.

Following previous literature (Husted et al. 2020), we choose three types of capital structure/leverage measurements using book value, namely, total debt ratio (TDR), which is the ratio of total debt to total assets, long-term debt ratio (LDR) which is the ratio of interest-bearing long-term debt to total assets and short-term debt ratio (SDR) which is the ratio of short-term debt to total assets. Using these different measures allows us to capture the influence of NIRP on firms' debt maturity structures.

We also employ various firm-specific variables, as in the literature. Firstly, the asset structure (AS) of firms—firms with more tangible assets have a higher liquidation value; hence, these firms are able to borrow against these assets, using them as collateral, at lower interest rates (Harris and Raviv, 1990; Titman and Wessels, 1988). Therefore, firms with higher collateralizable assets are more likely to take on more debt. Furthermore, firms with higher tangible assets are more likely to have stable returns and, therefore, can take on more debt (Daskalakis et al., 2017). On the other hand, agency costs associated with the scrutiny of highly levered firms may encourage firms

with less collateralisable assets to take on more debt. Earlier studies, such as that of Grossman and Hart (1982), suggest that due to the increased probability of bankruptcy, managers are less likely to consume excessive perquisites because bondholders or lenders will closely monitor the firm's activities (agency considerations). As a result, firms with less collateralised assets would want managers to be closely monitored and, hence, would take on more debt to withstand this scrutiny. On the other hand, given the assumption that debt maturity will be matched with asset life (Hall et al., 2004), firms with higher tangible assets would generally choose long-term debt at the expense of short-term debt, hence leading to a positive impact of asset structure on long-term debt but a negative impact on short-term debt. Hence, the literature on asset structure and firms' capital structure remains mixed. Asset structure is measured as the ratio of fixed assets to total assets.

Secondly, profitability (PROFIT) of firms—profitable firms can tap into their retained earning to finance their activities and hence are less likely to depend on debt, supporting the POT (Wu et al., 2022). Meanwhile, based on the TOT, more profitable firms can have greater borrowing capacity and, therefore, take on more debt (Fan et al., 2012; Fama and French, 2002). Profitability is measured as the ratio of earnings before interest and tax to total assets.

Thirdly, we include a measure of firm size (SIZE), indicating that larger firms are more likely to have stable earnings and to take on higher debt ratios (Titman and Wessels, 1988). Besides, smaller firms usually have less capacity to fully meet the credit requirements—particularly regarding information asymmetry—of banks or finance companies; hence, they are less likely to have higher debt ratios (Castanias, 1983). Firm size is measured as the natural logarithm of total assets.

Fourthly, firm liquidity (LIQ)—the ability of firms to meet short-term obligations, likely reduces the need for debt, especially short-term debt, as firms with ample cash can cover immediate needs without borrowing short-term debt, supporting the POT(De Jong et al., 2008). On the other hand, the TOT suggests that because bankruptcy costs are mostly associated with less liquid firms, these firms face greater obstacles in obtaining debt than more liquid firms (Degryse et al., 2012); hence, liquidity may have a positive impact on debt. Firm liquidity is measured as the ratio of cash to total assets.

Finally, firm age (AGE) is generally assumed to be a going concern. Hence, firms that have been in the market for a long time will likely have built a reputation over the years (Diamond, 1989). Therefore, compared to younger firms, older firms are likely to have higher debt ratios, given that they are more likely to have access to debt at potentially cheaper rates (Abor and Biekpe, 2009; Petersen and Rajan, 1994). Firm age is measured as the number of years since the firm was incorporated.

Independent Variables: macroeconomic variables

We also control for macroeconomic variables identified in the literature as affecting firm capital structure. Firstly, Gross Domestic Product (GDP) growth is used as a measure of aggregate demand in the economy. Higher growth would suggest higher demand for goods and services; hence, firms are more likely to increase their revenue and profits, leading to more retained earnings and, consequently, a reduced need for debt. On the other hand, the higher demand for goods and services could lead firms to take on more debt to expand and meet the growing demand. These views are supported by studies such as Frank and Goyal (2009).

Secondly, inflation (INF)—when firms expect inflation to increase, they are likely to increase their debt, given that debt will be relatively cheaper to pay when inflation rises (Frank and Goyal, 2009). Following Graham et al. (2015), we use the percentage change in the consumer price index as a measure of expected inflation.

3.1 Descriptive statistics

Table 1 presents summary statistics for the key variables in the sample, covering 472,778 firm-year observations. On average, firms exhibit a total debt ratio (TDR) of 0.48, indicating that debt finances nearly half of their capital structure. Interestingly, the average short-term debt ratio (SDR) is considerably higher (0.30) than the long-term debt ratio (LDR) (0.13), suggesting a general reliance on short-term debt instruments. The distribution of firm size (proxied by the natural logarithm of total assets) spans a wide range, with a median of 18.21. On the other hand, firm age has a mean of 26 years, with some firms as old as 123 years. Profitability, measured as return on assets, is low on average (-0.02), and the liquidity ratio shows substantial heterogeneity across firms.

Table 2 disaggregates these statistics by treatment status. Firms in NIRP-exposed economies (treated group) exhibit markedly higher total debt ratios (mean = 0.58) compared to the control group (mean = 0.48), with the median also significantly higher at 0.59. This pattern is echoed in the LDR metric, where treated firms have a mean of 0.16 and a median of 0.14, compared to 0.12 and 0.06 in the control group, respectively. These patterns are consistent with a debt maturity extension among treated firms. Meanwhile, SDR remains largely comparable across the groups, albeit marginally higher in the treated group.

These patterns are visually confirmed in the kernel density plots in Figure 2. The distribution of total debt ratios for treated firms (in blue) is noticeably shifted to the right relative to that of non-treated firms (in red), consistent with the higher central tendency observed in the summary statistics. Similarly, the LDR distribution for treated firms is more right-skewed, reflecting greater long-term debt holdings. In contrast, the SDR distributions overlap substantially, rein-

forcing the interpretation that while short-term debt levels remain relatively stable, the shift in capital structure under NIRP is primarily driven by an increase in long-term debt. These visual patterns complement the tabulated statistics and provide additional descriptive support for the hypothesis that firms exposed to NIRP reallocate their debt composition in favour of long-term instruments, potentially to lock in persistently low rates. This motivates a deeper investigation using causal identification strategies.

Table 1: Descriptive Statistics

Variable	Obs.	Q1	Median	Mean	Q3	Max
TDR	472,778	0.31	0.49	0.48	0.66	0.96
SDR	472,778	0.15	0.27	0.30	0.43	0.84
LDR	472,778	0.01	0.07	0.13	0.20	0.65
AGE	472,778	10.00	19.00	26.02	33.00	123.00
PROFIT	472,778	-0.03	0.04	-0.02	0.09	0.35
LIQ	472,778	0.03	0.08	0.16	0.20	0.95
SIZE	472,778	16.64	18.21	18.12	19.85	24.07
AS	472,778	0.32	0.51	0.51	0.70	0.98
INF	472,778	1.53	2.37	3.14	3.77	15.84
GDP growth	472,778	1.84	3.09	3.53	5.95	11.39

Notes: TDR: Total debt ratio; SDR: Short-term debt ratio; LDR: Long-term debt; AGE: Age of firms; PROFIT: return on assets; LIQ: ratio of liquid assets to total assets; SIZE: natural logarithm of total assets; INF: Inflation measured percentage change in consumer prices (%); GDP growth: Real GDP growth (%).

Table 2: Summary Statistics of Debt Ratios by Treatment Group

Variable	Treatment	N	Q1	Median	Mean	Q3	Max
TDR	Control	435,235	0.30	0.48	0.48	0.65	0.96
TDR	Treated	37,543	0.45	0.59	0.58	0.72	0.96
SDR	Control	435,235	0.14	0.26	0.30	0.42	0.84
SDR	Treated	37,543	0.21	0.31	0.33	0.45	0.84
LDR	Control	435,235	0.00	0.06	0.12	0.19	0.65
LDR	Treated	37543	0.05	0.14	0.16	0.24	0.65

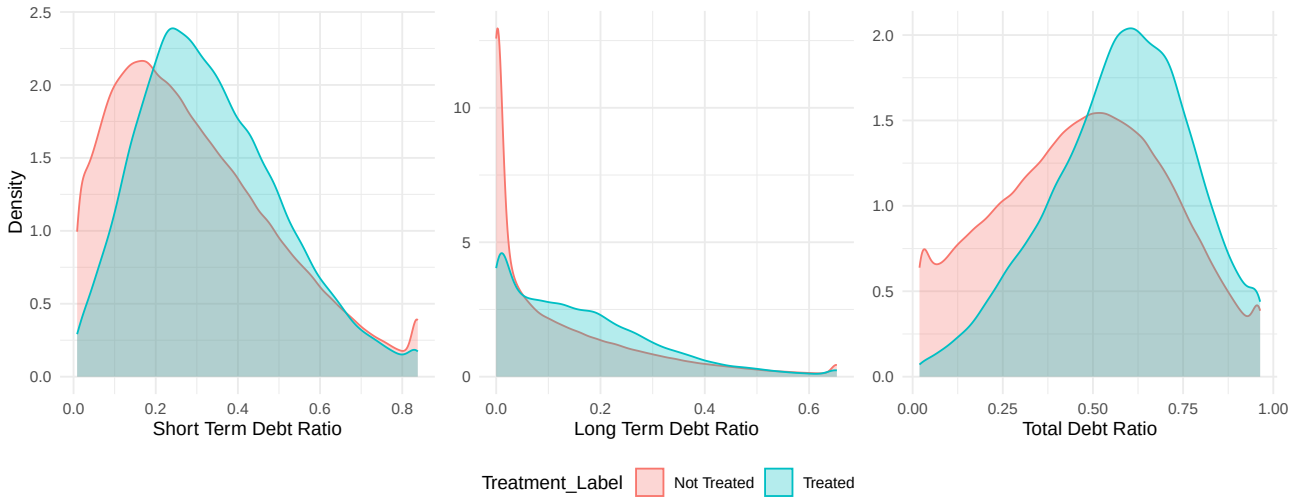

Figure 2: Density Plots of Debt Ratios

4 Empirical Methodology

4.1 Overview of ECB's UMP Decisions and Identification of NIRP

Tables B.1 and B.2[3] summarise the major unconventional monetary policy actions undertaken by the European Central Bank (ECB) from 2013 onward. These include the introduction of the Negative Interest Rate Policy (NIRP), the Asset Purchase Programme (APP) and the Pandemic Emergency Purchase Programme (PEPP)—both Quantitative Easing (QE) programs—and the Forward Guidance (FG) framework. NIRP was first implemented in June 2014 through successive reductions of the deposit facility rate into negative territory, marking a regime shift in the ECB's operational toolkit. The APP, launched in 2014 and expanded in 2015, represented the ECB's quantitative-easing effort to stabilise the term structure and support private-sector credit. Forward guidance, introduced in 2013, complemented these tools by anchoring expectations about the persistence of low interest rates. Finally, PEPP, introduced in March 2020, represented an extraordinary response to the COVID-19 shock, expanding the scale and flexibility of asset purchases.

The identification of the causal effect of NIRP on corporate financing behaviour is complicated by the concurrent implementation of both QE and FG (Tables B.1 and B.2). We estimate a Difference-in-Differences (DiD) design that exploits cross-country and temporal variation in exposure, with euro area firms treated as the treatment group and non–euro area firms as controls. We control for balance-sheet policies (APP and PEPP), and the FG Frameworks directly with

[3]See Rostagno et al. (2021) for an in-depth discussion on the unconventional monetary policy decisions.

country–year dummy variables when the respective programme is active, and zero otherwise. Alongside macro country–year controls (real GDP growth and inflation), these dummies net out policy-liquidity conditions unrelated to NIRP, so the DiD coefficient captures firms' *demand-side* adjustment in SDR, LDR, and TDR to the negative policy rate. Firm-level controls (age, profitability, liquidity, size, and asset structure) absorb heterogeneity in financing needs. Under these controls and fixed effects, the DiD coefficient is interpreted as the *demand-side* response of firms' debt choices (SDR, LDR, TDR) to the pricing channel of monetary accommodation—namely, the negative policy rate—holding constant contemporaneous asset-purchase and forward guidance components. We also assess the parallel-trends assumption using pre-treatment leads and event-study graphs.

Parallel Trends

To ensure the validity of the parallel trends assumption in our Difference-in-Differences (DiD) framework, we examine the dynamics of the firm-level debt measures—short-term, long-term, and total debt ratios—comparing treated firms in the euro area with control firms outside the euro area. The identifying assumption is that, in the absence of the ECB's introduction of the negative interest rate policy (NIRP) in 2014, the debt trajectories of treated and control firms would have evolved similarly over time. Following from other studies (He and Wang, 2017), we formally evaluate this assumption using an event-study specification that includes multiple pre-treatment leads (four years prior to treatment), setting the year immediately before treatment ($t = -1$) as the reference period. Visual inspection of pre-treatment trends, as shown in Figure 3 discussed later under subsection 5.7, indicates that the coefficients on pre-treatment leads are statistically indistinguishable from those in the base period (the year preceding the NIRP implementation). This visual inspection suggests a plausible causal effect and parallel trends.

Beyond the event–study diagnostics, we strengthen the credibility of parallel trends using the panel-data matching framework of Imai et al. (2023), which we describe in our robustness analysis. For each treated firm–year, we construct a matched risk set of control firm–years drawn from the same calendar year that share an identical treatment history over a pre-specified lead and lag window. We then estimate average treatment effects via difference-in-differences within the matched sets. This design-based procedure clarifies the counterfactual construction, reduces model dependence relative to two-way fixed-effects regression, and provides transparent balance diagnostics. As we discuss further later on, applied to all three debt outcomes (*SDR, LDR, TDR*), the matched DiD estimates are consistent with our main findings.

4.2 Model Specification

Following from the literature (Carbó-Valverde et al., 2021), we identify the causal effect of NIRP on the capital structure using the Difference-in-Differences (DiD) framework.

$$DR_{ict} = \theta_i + \eta_t + \lambda_{ht} + \beta(Treated_i \times POST_t) + \boldsymbol{\gamma X}_{ict} + \varepsilon_{ict} \tag{1}$$

where i,c,t represents firm i in country c at time t. DR is the measure of the capital structure/financing decision of the firms proxied by the ratio of long-term debt to total assets (LDR), the ratio of short-term debt to total assets (SDR), and the ratio of total debt to total assets (TDR).

Treated is a dummy variable equal to 1 for firms in a country (for countries in the euro area) that adopted an NIRP and 0 otherwise. POST is a dummy variable equal to 1 in the year (2014) of NIRP adoption and afterwards, and 0 otherwise. Essentially, the interaction of the two represents our difference-in-differences variable (DiD hereafter). The ECB has cut the Deposit Facility Rate(DFR) in June 2014 from 0 to -0.1%, from -0.1% to -0.2% in September 2014, from -0.2% to -0.3% in December 2015, from -0.3% to -0.4% in March 2016, and from -0.4% to -0.5% in September 2019 and remained so until it was changed to 0 in July 2022[4]. We also add the quantitative easing (ECB_QE) and forward guidance (ECB_FG) dummies for the euro area.

β is the coefficient of interest, which is the average treatment effect, indicating the average difference in capital structure ($DEBT$) between the treated and control groups before and after the implementation of the negative interest rate policy. A positive and significant estimate of β would indicate that firms in countries with NIRP exhibit higher leverage after implementing the NIRP, provided that capital structure decisions and other shocks evolve similarly for both treated and control groups, as discussed under the identification strategy earlier.

X_{ikt} is a vector of firm-specific and macroeconomic control variables identified in the literature to influence firms' capital structure. These include firm-specific controls, such as age, profitability, liquidity, size, and asset structure, as well as macroeconomic variables, including inflation and real GDP growth. $\boldsymbol{\gamma}$ is a vector of parameters for the control variables; θ_i and η_t are the firm and year fixed effects, respectively; λ_{ht} is the industry (2-digit SIC)-year fixed effects to capture any shocks in the industry of the firms over time that may affect firms' financing decision; $\varepsilon_{i,k,t}$ is the idiosyncratic error term.

We estimate this model using the traditional Two-way fixed effects (TWFE) approach. Given the 2×2-DiD framework with no heterogeneity in treatment effects across either time or units, the standard within estimator produces consistent estimates (Dube et al., 2023; Roth et al.,

[4] https://www.ecb.europa.eu/stats/policy_and_exchange_rates/key_ecb_interest_rates/html/index.en.html

2023).

5 Empirical Results

In this section, we discuss the results from our difference-in-differences estimations. First, the results are presented in Tables 3 to 5. Across all debt measures—total (TDR), long-term (LDR), and short-term (SDR)—the coefficient on the DiD variable, representing the impact of the Negative Interest Rate Policy (NIRP), exhibits a clear pattern of heterogeneous effects across debt maturities.

Table 3 shows the results when *TDR* is used as the dependent variable. From the table, we see that the coefficient of *DiD* (*i.e.* $Treated_i \times POST_t$) was insignificant in the full model, indicating that, on average, the introduction of negative rates did not significantly alter firms' overall leverage. This likely reflects offsetting dynamics across debt maturities. On the other hand, from Table 4, when *SDR* is used as the dependent variable, the coefficient of *DiD* is negative and statistically significant at the 1% significance level. This indicates that firms in a negative interest rate environment tend to reduce their short-term debt. On average, this decrease was around 1.42%. Meanwhile, from Table 5, the *DiD* indicator has a statistically positive impact on the long-term debt ratio at 1% significance level. The average increase was about 1.57%.

Meanwhile, quantitative easing (ECB_QE) appears to reinforce this shift, with negative coefficients on SDR and positive on LDR, as purchases lower long yields and enhance supply. On the other hand, forward guidance (ECB_FG)—which generally communicates the ECB's commitment to keeping rates low for an extended period—enters with a negative sign for all debt measures. This suggests that while firms anticipate persistently low rates, they respond with greater prudence rather than aggressive borrowing, consistent with a stabilisation role of FG. Forward guidance, therefore, moderates the short-term effects of NIRP by anchoring expectations, leading to a smoother and more gradual adjustment in debt maturity and leverage. Thus, in the presence of other unconventional monetary polices (QE and FG), the NIRP continues to capture an independent, maturity-specific response rather than a conflation of policy channels.

These results suggest that firms in the euro area adopted a term maturity restructuring of their debt portfolio in order to benefit from the negative interest rate policy. This supports the "borrow long, dump short" mechanism where NIRP encourages firms to extend maturities. This may support the theory of *debt market-timing*, where firms adjust their capital structure in response to market conditions, such as the NIRP, to take advantage of lower interest rates (Bolton et al., 2013; Karpavičius and Yu, 2017). Moreover, as we have demonstrated in our theoretical model, decreasing short-term debt to increase long-term debt may be due to the higher leverage adjustment cost associated with short-term debt compared to the higher risk

premium on long-term debt. This supports the argument of Jungherr and Schott (2021), who find that long-term debt saves rollover costs.

While it is likely that short-term debt has lower initial issuance costs, such as legal and banking fees, because it's simpler to arrange, the need to refinance short-term debt frequently—potentially every year—can lead to higher total adjustment costs over time. Long-term debt, although it incurs higher upfront costs, benefits from stability, with lower annual amortisation expenses for debt issuance costs. An unexpected detail is that in negative interest rate environments, firms may shift toward long-term debt to lock in low rates and avoid the risks associated with frequent refinancing, aligning with our hypothesis that firms reduce short-term debt and increase long-term debt, potentially due to the higher refinancing costs of short-term debt. This hypothesis aligns with the argument of Wu et al. (2022), who show that short-term borrowing makes firms vulnerable to credit denial, may force them to curtail operations or fire-sale assets, and increases the opportunity costs of management due to the need for more frequent debt issuance.

Impact of control variables

Here, we discuss the impact of our control variables on firms' capital structure. The results indicate that firm age generally has a positive impact on firms' total debt ratios. On the other hand, while firm age has a positive impact on short-term debt in all estimations at the 1% significance level, it has a negative impact on long-term debt. These results suggest that, compared to younger firms, older firms seem to take on more short-term debt while reducing long-term debt. While current studies (Abor and Biekpe, 2009; Petersen and Rajan, 1994; Diamond, 1989) show that older firms are more likely to take on more debt due to their reputation, our findings show that this may be driven by short-term debt. This could be because, as firms age, they often operate more stably and require more short-term debt to cover daily expenses, such as payroll and inventory, thereby increasing their total debt. Older firms may also have established relationships with banks, making it easier to obtain short-term working capital loans. Mature firms typically need less long-term debt because they are not expanding as rapidly as younger firms. They might pay off long-term loans or avoid taking on new ones to maintain financial flexibility, focusing instead on short-term financing for operational needs. Additionally, we observe that firm profitability negatively affects all measures of debt at the 1% significance level, supporting the *POT*. As mentioned earlier, profitable firms can have more retained earnings to draw upon and, therefore, may be less dependent on debt, especially short-term debt, compared to long-term debt.

Meanwhile, more liquid firms require less debt, with liquidity having the most pronounced negative impact on short-term debt. This is intuitive because the ability to meet short-term obligations likely reduces the need for short-term debt, as firms with ample cash can cover

Table 3: NIRP and Total Debt – Full Sample

Model:	(1)	(2)	(3)	(4)
DiD	-0.0044	-0.0033	0.0044	0.0042
	(0.0032)	(0.0033)	(0.0031)	(0.0031)
ECB_QE	0.0096***	0.0089***	0.0064**	0.0070**
	(0.0031)	(0.0031)	(0.0030)	(0.0030)
ECB_FG	-0.0191***	-0.0159***	-0.0172***	-0.0154***
	(0.0024)	(0.0024)	(0.0023)	(0.0023)
AGE			0.0022***	0.0015*
			(0.0008)	(0.0008)
PROFIT			-0.1352***	-0.1350***
			(0.0023)	(0.0023)
LIQ			-0.4180***	-0.4142***
			(0.0050)	(0.0050)
SIZE			-7.18×10^{-5}	0.0001
			(0.0008)	(0.0008)
AS			-0.1483***	-0.1502***
			(0.0048)	(0.0048)
INF			0.0022***	0.0018***
			(0.0002)	(0.0002)
GDP growth			0.0022***	0.0020***
			(0.0002)	(0.0002)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	No	Yes	No	Yes
No. of Firms	58,418	58,418	58,418	58,418
Observations	472,778	472,778	472,778	472,778
R^2	0.69	0.69	0.72	0.72

Notes: This table provides the estimates of the difference-in-difference estimates of the impact of NIRP on capital structure using total debt ratio (TDR) following Equation 1 with Two-way Fixed Effects (TWFE), where DiD is $Treated_i \times POST_t$. *Clustered (firmid) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.*

immediate needs without borrowing in the short term, also supporting the *POT* (Wu et al., 2022; De Jong et al., 2008). From the tables, we see that firm size has a positive impact on long-term debt but a negative impact on short-term debt. This supports studies, such as that of Hall et al. (2004), which find that firm size has a negative impact on short-term debt. In contrast, other studies, including Degryse et al. (2012), find a positive impact on long-term debt. Larger firms can often leverage high-value assets to borrow more long-term debt because they are less likely to go bankrupt, suggesting their ability to repay their debt (Ang et al., 1982; Warner, 1977).

Additionally, the results indicate that firms with larger asset capacity are more likely to take on long-term debt compared to short-term debt. Therefore, asset structure has a negative

Table 4: NIRP and Short-Term Debt – Full Sample

Model:	(1)	(2)	(3)	(4)
DiD	-0.0180***	-0.0166***	-0.0112***	-0.0108***
	(0.0026)	(0.0026)	(0.0024)	(0.0024)
ECB_QE	-0.0007	-0.0005	-0.0041*	-0.0039*
	(0.0025)	(0.0025)	(0.0023)	(0.0024)
ECB_FG	-0.0077***	-0.0066***	-0.0076***	-0.0067***
	(0.0019)	(0.0019)	(0.0017)	(0.0017)
AGE			0.0019***	0.0017***
			(0.0005)	(0.0006)
PROFIT			-0.0998***	-0.1010***
			(0.0018)	(0.0018)
LIQ			-0.4114***	-0.4100***
			(0.0042)	(0.0042)
SIZE			-0.0170***	-0.0165***
			(0.0006)	(0.0006)
AS			-0.3075***	-0.3065***
			(0.0041)	(0.0041)
INF			0.0024***	0.0022***
			(0.0001)	(0.0001)
GDP growth			0.0009***	0.0009***
			(0.0001)	(0.0001)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	No	Yes	No	Yes
Observations	472,778	472,778	472,778	472,778
No. of Firms	58,418	58,418	58,418	58,418
R^2	0.68	0.68	0.73	0.73

Notes: This table provides the estimates of the difference-in-difference estimates of the impact of NIRP on capital structure using short-term debt ratio (SDR) following Equation 1 with Two-way Fixed Effects (TWFE), where DiD is $Treated_i \times POST_t$. *Clustered (firmid) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.*

Table 5: NIRP and Long-Term Debt – Full Sample

Model:	(1)	(2)	(3)	(4)
DiD	0.0153***	0.0146***	0.0168***	0.0160***
	(0.0024)	(0.0025)	(0.0024)	(0.0024)
ECB_QE	0.0146***	0.0138***	0.0148***	0.0149***
	(0.0024)	(0.0025)	(0.0024)	(0.0024)
ECB_FG	-0.0084***	-0.0069***	-0.0069***	-0.0063***
	(0.0019)	(0.0019)	(0.0018)	(0.0019)
AGE			-0.0010**	-0.0013**
			(0.0005)	(0.0006)
PROFIT			-0.0328***	-0.0317***
			(0.0012)	(0.0012)
LIQ			0.0021	0.0043
			(0.0030)	(0.0030)
SIZE			0.0149***	0.0147***
			(0.0005)	(0.0005)
AS			0.1201***	0.1180***
			(0.0032)	(0.0032)
INF			-0.0001	-0.0003*
			(0.0001)	(0.0001)
GDP growth			0.0009***	0.0007***
			(0.0001)	(0.0001)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	No	Yes	No	Yes
No. of Firms	58,418	58,418	58,418	58,418
Observations	472,778	472,778	472,778	472,778
R^2	0.64	0.65	0.65	0.66

Notes: This table provides the estimates of the difference-in-difference estimates of the impact of NIRP on capital structure using long-term debt ratio (TDR) following Equation 1 with Two-way Fixed Effects (TWFE), where DiD is $Treated_i \times POST_t$. *Clustered (firmid) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.*

impact on both short-term and total debt ratios, while having a positive impact on long-term debt. This is in line with studies like (Hall et al., 2004; Myers, 1977), who note that tangible assets reduce short-term debt, as they are better suited for long-term financing, increasing long-term debt. This is because the proportion of tangible to intangible assets affects collateral availability; hence, firms with more tangible assets can secure more long-term debt. What we observe is that, on aggregate, the reduction in short-term debt exceeds the increase in long-term debt; hence, the total debt ratio declines.

At the macro level, we find that overall, inflation has a positive impact on short-term debt but a marginally negative impact on long-term debt. Indeed, inflation impacts short-term and long-term debt differently due to its effects on borrowing costs, repayment dynamics, and firm behaviour. When inflation rises, firms may prefer short-term debt to maintain flexibility and financial stability. Short-term debt enables firms to refinance frequently, allowing them to adjust to new interest rates as inflation fluctuates. This avoids locking into rates that might not keep pace with inflation, which could increase real borrowing costs if rates lag behind. On the other hand, high inflation makes long-term debt less appealing because it locks firms into fixed nominal rates for several years, such as 5 or 10. If inflation outpaces these rates, the real value of debt payments declines, initially benefiting borrowers, but lenders, anticipating this, demand higher rates or shorter terms, thereby raising the cost of long-term debt. Therefore, the decrease in long-term debt may result from lenders' preference for short-term debt or for firms seeking to keep up with changes in inflation. Additionally, GDP growth has a positive impact on all types of debt. This suggests that higher aggregate demand in the economy may require higher investments; hence, firms may borrow more to finance these investments to meet the growing demand.

5.1 Robustness: Controlling for Bank Credit Supply

To strengthen our identification of NIRP and to reinforce the demand-side channel of NIRP on corporate debt maturity, we account for potential supply-side confounding from credit supply. We add the lagged bank credit growth[5] as an additional control. This allows us to control for time-varying bank supply-side effects. Bank credit growth at $t-1$ serves as a broader proxy for prior credit supply dynamics, capturing aggregate lending expansions that could influence firm borrowing independently of NIRP. The DiD coefficient on NIRP thus isolates the policy's demand-side impact—firms' preference to extend maturities due to heightened short-term refinancing costs—net of these supply factors.

The results are presented in Tables 6 . When bank credit growth is introduced as a direct proxy for aggregate credit supply conditions, it enters with economically intuitive and highly

[5]Based on credit provided to the private sector by banks. Data is from World Development Indicators.

robust signs across all specifications. In both parsimonious and fully controlled regressions, stronger credit growth is associated with higher long-term debt ratios, lower short-term debt ratios, and lower total debt ratios. This pattern is what one would expect if banks, when flush with lending capacity, preferentially steer firms toward longer maturities — either because long-term loans are more profitable under the prevailing yield curve or because operations such as Targeted Longer-Term Refinancing Operations (TLTROs) (which we discuss in the next section) explicitly reward longer-duration lending. The negative effect on total debt is consistent with credit supply expansions occurring predominantly through maturity extension rather than a broad-based increase in leverage.

The NIRP (DiD) coefficient tells a similarly coherent story and, crucially, *survives the inclusion of credit supply with virtually no attenuation*. In specifications without firm-level controls, NIRP induces a statistically significant maturity shift — positive on LDR, negative on SDR — with no discernible impact on total debt. Once observable firm characteristics are accounted for, the NIRP effect strengthens further, and notably, becomes positive and statistically significant for total debt. Thus, conditioning on these controls uncovers that NIRP does, in fact, spur a modest but meaningful increase in overall indebtedness, in addition to its maturity-rebalancing effect.

Taken together, these results deliver a sharp separation of demand- and supply-side forces. Bank credit growth captures a clear *supply-driven maturity-extension channel* — when banks are eager to lend, they push credit toward longer horizons. NIRP, by contrast, operates primarily through *firm demand*: even after absorbing the full influence of credit supply fluctuations, negative policy rates induce firms to actively restructure their liabilities, reducing short-term exposure and increasing both long-term and total debt. This evidence establishes debt-maturity restructuring and moderate leverage expansion as two distinct, demand-driven margins through which negative interest rates are transmitted to the real economy.

5.2 Robustness: Controlling for Targeted Longer-Term Refinancing Operations (TLTRO)

To further mitigate the potential confounding effects of other unconventional monetary policy measures and to further isolate the demand-side channel of NIRP on corporate debt maturity, we extend our baseline specification to include the *Targeted Longer-Term Refinancing Operations* (TLTROs). The TLTROs—first announced on 5 June 2014 (TLTRO-I), recalibrated on 10 March 2016 (TLTRO-II), and followed by a third series on 7 March 2019 (TLTRO-III)—were Eurosystem operations providing long-term central bank credit to euro area banks at attractive conditions. In this setting, the NIRP coefficient identifies the price and expectations component

of monetary transmission net of contemporaneous targeted liquidity provision. The inclusion of TLTRO effects then reveals how targeted funding reshapes debt maturity and leverage, clarifying whether TLTRO amplifies or attenuates the maturity-rebalancing mechanism of NIRP.

The results reported in Tables 9–11 show that the main conclusion about NIRP's maturity rebalancing effect remains robust. The coefficient on the difference-in-differences estimator (DiD) retains its sign across all specifications—negative for short-term debt and positive for long-term debt—confirming that firms respond to negative rates by extending debt maturities. However, the magnitude of these effects diminishes once TLTRO and credit growth are taken into account, implying that part of the NIRP transmission operates through improved bank liquidity and credit provision. The positive and statistically significant coefficients on ECB_TLTRO for total and long-term debt indicate that these targeted liquidity injections facilitated credit expansion and supported long-term borrowing, thereby amplifying the maturity-extension channel of NIRP. By contrast, the modest negative effect of TLTRO on short-term debt suggests a substitution away from short-term borrowing when targeted funding is available.

Lagged bank credit growth exhibits heterogeneous effects across debt maturities, being negative for short-term debt but positive for long-term debt. This implies that greater lending activity allows firms to refinance and lock in longer-term obligations while reducing rollover exposure. Overall, accounting for these channels strengthens the causal interpretation of our results and helps to isolate the effect of NIRP from contemporaneous unconventional policy interventions.

5.3 Robustness: Interaction with Bank Credit Supply

We extend the empirical specification by introducing an interaction term between the NIRP treatment variable (DiD) and lagged bank credit growth to examine whether the transmission of negative policy rates depends on prevailing credit-supply conditions. This interaction captures whether NIRP operates as a *complementary* channel—amplifying the effects of an expanding credit supply—or as a *substitute* that compensates for weak lending activity.

The results reported in Tables 12–14 display a clear asymmetry across debt maturities. The interaction term ($DiD \times Bank\ Credit\ Growth_{t-1}$) is positive and statistically significant for short-term debt, indicating that when bank credit expands, firms' short-term borrowing becomes more responsive to NIRP. In this context, negative rates and credit growth operate as *complementary* forces that jointly reinforce liquidity-driven balance-sheet adjustments.

By contrast, the interaction coefficients for total and long-term debt are small and generally insignificant in some specifications, but the dominant pattern is one of substitution: NIRP appears to *substitute* for weak credit supply—its influence on long-term financing is more pronounced when lending conditions are tight and firms face higher refinancing constraints.

Table 6: NIRP and Total Debt, Controlling for Credit Supply – Full Sample

Model:	(1)	(2)	(3)	(4)
ECB_NIRP	0.0045	0.0035	0.0134***	0.0124***
	(0.0032)	(0.0032)	(0.0030)	(0.0031)
ECB_QE	0.0008	-0.0020	0.0028	0.0010
	(0.0031)	(0.0032)	(0.0030)	(0.0030)
ECB_FG	-0.0178***	-0.0164***	-0.0170***	-0.0161***
	(0.0024)	(0.0024)	(0.0023)	(0.0023)
Bank Credit Growth(t-1)	-0.0335***	-0.0322***	-0.0338***	-0.0342***
	(0.0048)	(0.0048)	(0.0046)	(0.0046)
AGE			0.0021**	0.0014
			(0.0009)	(0.0009)
PROFIT			-0.1889***	-0.1867***
			(0.0037)	(0.0037)
LIQ			-0.4151***	-0.4134***
			(0.0066)	(0.0066)
SIZE			0.0185***	0.0185***
			(0.0012)	(0.0013)
AS			-0.1267***	-0.1289***
			(0.0062)	(0.0062)
INF			0.0016***	0.0013***
			(0.0002)	(0.0002)
GDP growth			0.0018***	0.0015***
			(0.0002)	(0.0002)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	No	Yes	No	Yes
No. of Firms	42,967	42,967	42,967	42,967
Observations	300,349	300,349	300,349	300,349
R^2	0.75	0.75	0.78	0.78

Notes: This table provides the estimates of the difference-in-difference estimates of the impact of NIRP on capital structure using total debt ratio (TDR) **controlling for bank credit supply** following Equation 1 with Two-way Fixed Effects (TWFE), where DiD is $Treated_i \times POST_t$. *Clustered (firmid) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.*

Table 7: NIRP and Short-Term Debt, Controlling for Credit Supply – Full Sample

Model:	(1)	(2)	(3)	(4)
ECB_NIRP	-0.0161***	-0.0156***	-0.0106***	-0.0103***
	(0.0026)	(0.0026)	(0.0024)	(0.0024)
ECB_QE	-0.0051**	-0.0060**	-0.0048*	-0.0057**
	(0.0026)	(0.0026)	(0.0024)	(0.0025)
ECB_FG	-0.0071***	-0.0064***	-0.0081***	-0.0072***
	(0.0019)	(0.0019)	(0.0018)	(0.0018)
Bank Credit Growth(t-1)	-0.0650***	-0.0647***	-0.0601***	-0.0613***
	(0.0041)	(0.0042)	(0.0039)	(0.0040)
AGE			0.0015***	0.0014**
			(0.0006)	(0.0006)
PROFIT			-0.1267***	-0.1278***
			(0.0028)	(0.0028)
LIQ			-0.4008***	-0.4007***
			(0.0054)	(0.0054)
SIZE			-0.0096***	-0.0093***
			(0.0009)	(0.0010)
AS			-0.3233***	-0.3220***
			(0.0053)	(0.0053)
INF			0.0017***	0.0017***
			(0.0002)	(0.0002)
GDP growth			0.0002	0.0002*
			(0.0001)	(0.0001)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	No	Yes	No	Yes
No. of Firms	42,967	42,967	42,967	42,967
Observations	300,349	300,349	300,349	300,349
R^2	0.74	0.75	0.78	0.78

Notes: This table provides the estimates of the difference-in-difference estimates of the impact of NIRP on capital structure using short-term debt ratio (SDR) **controlling for bank credit supply** following Equation 1 with Two-way Fixed Effects (TWFE), where DiD is $Treated_i \times POST_t$. *Clustered (firmid) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.*

Table 8: NIRP and Long-Term Debt, Controlling for Credit Supply – Full Sample

Model:	(1)	(2)	(3)	(4)
ECB_NIRP	0.0218***	0.0205***	0.0249***	0.0239***
	(0.0025)	(0.0025)	(0.0025)	(0.0025)
ECB_QE	0.0098***	0.0081***	0.0112***	0.0102***
	(0.0026)	(0.0026)	(0.0025)	(0.0026)
ECB_FG	-0.0080***	-0.0079***	-0.0064***	-0.0068***
	(0.0020)	(0.0020)	(0.0019)	(0.0020)
Bank Credit Growth(t-1)	0.0532***	0.0545***	0.0461***	0.0473***
	(0.0038)	(0.0038)	(0.0037)	(0.0038)
AGE			-0.0009	-0.0013**
			(0.0006)	(0.0006)
PROFIT			-0.0542***	-0.0513***
			(0.0021)	(0.0021)
LIQ			-0.0071*	-0.0054
			(0.0043)	(0.0042)
SIZE			0.0264***	0.0263***
			(0.0009)	(0.0009)
AS			0.1487***	0.1467***
			(0.0045)	(0.0045)
INF			0.0001	-7.75×10^{-6}
			(0.0002)	(0.0002)
GDP growth			0.0011***	0.0008***
			(0.0001)	(0.0001)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	No	Yes	No	Yes
No. of Firms	42,967	42,967	42,967	42,967
Observations	300,349	300,349	300,349	300,349
R^2	0.68	0.69	0.70	0.71

Notes: This table provides the estimates of the difference-in-difference estimates of the impact of NIRP on capital structure using long-term debt ratio (TDR) **controlling for bank credit supply** following Equation 1 with Two-way Fixed Effects (TWFE), where DiD is $Treated_i \times POST_t$. *Clustered (firmid) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.*

Table 9: NIRP and Total Debt with TLTRO and Credit Growth – Full Sample

Model:	(1)	(2)	(3)	(4)
DiD	-0.0050	-0.0045	0.0053	0.0054*
	(0.0034)	(0.0034)	(0.0032)	(0.0032)
ECB_QE	0.0116***	0.0117***	0.0074**	0.0048
	(0.0033)	(0.0033)	(0.0033)	(0.0033)
ECB_FG	-0.0164***	-0.0147***	-0.0165***	-0.0156***
	(0.0023)	(0.0023)	(0.0023)	(0.0023)
ECB_TLTRO	0.0087***	0.0080***	0.0076***	0.0065***
	(0.0015)	(0.0015)	(0.0015)	(0.0015)
AGE	0.0022***	0.0015*	0.0021**	0.0014
	(0.0008)	(0.0008)	(0.0009)	(0.0009)
PROFIT	-0.1352***	-0.1350***	-0.1889***	-0.1867***
	(0.0023)	(0.0023)	(0.0037)	(0.0037)
LIQ	-0.4181***	-0.4142***	-0.4151***	-0.4134***
	(0.0050)	(0.0050)	(0.0066)	(0.0066)
SIZE	-7.05×10^{-5}	0.0001	0.0185***	0.0185***
	(0.0008)	(0.0008)	(0.0012)	(0.0013)
AS	-0.1484***	-0.1502***	-0.1268***	-0.1290***
	(0.0048)	(0.0048)	(0.0062)	(0.0062)
INF	0.0022***	0.0018***	0.0016***	0.0014***
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
GDP growth	0.0022***	0.0020***	0.0018***	0.0015***
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
Bank Credit Growth(t-1)			-0.0335***	-0.0339***
			(0.0046)	(0.0046)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	No	Yes	No	Yes
No. of Firms	58,418	58,418	42,967	42,967
Observations	472,778	472,778	300,349	300,349
R^2	0.72	0.72	0.78	0.78

Notes: This table provides the estimates of the difference-in-difference estimates of the impact of NIRP on capital structure using total debt ratio (TDR) following Equation 1 with Two-way Fixed Effects (TWFE), where DiD is $Treated_i \times POST_t$. *Clustered (firmid) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.*

Table 10: NIRP and Short-Term Debt TLTRO and Credit Growth – Full Sample

Model:	(1)	(2)	(3)	(4)
DiD	-0.0082***	-0.0079***	-0.0047*	-0.0041
	(0.0027)	(0.0028)	(0.0027)	(0.0028)
ECB_QE	-0.0058**	-0.0055**	-0.0080***	-0.0092***
	(0.0026)	(0.0026)	(0.0027)	(0.0027)
ECB_FG	-0.0079***	-0.0070***	-0.0085***	-0.0076***
	(0.0017)	(0.0017)	(0.0017)	(0.0018)
ECB_TLTRO	-0.0028**	-0.0027*	-0.0055***	-0.0058***
	(0.0014)	(0.0014)	(0.0014)	(0.0014)
AGE	0.0019***	0.0017***	0.0015***	0.0014**
	(0.0005)	(0.0006)	(0.0006)	(0.0006)
PROFIT	-0.0998***	-0.1010***	-0.1267***	-0.1278***
	(0.0018)	(0.0018)	(0.0028)	(0.0028)
LIQ	-0.4114***	-0.4100***	-0.4007***	-0.4006***
	(0.0042)	(0.0042)	(0.0054)	(0.0054)
SIZE	-0.0171***	-0.0165***	-0.0096***	-0.0093***
	(0.0006)	(0.0006)	(0.0009)	(0.0010)
AS	-0.3074***	-0.3065***	-0.3232***	-0.3219***
	(0.0041)	(0.0041)	(0.0053)	(0.0053)
INF	0.0024***	0.0022***	0.0017***	0.0016***
	(0.0001)	(0.0001)	(0.0002)	(0.0002)
GDP growth	0.0009***	0.0009***	0.0001	0.0002*
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Bank Credit Growth(t-1)			-0.0604***	-0.0616***
			(0.0039)	(0.0040)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	No	Yes	No	Yes
No. of Firms	58,418	58,418	42,967	42,967
Observations	472,778	472,778	300,349	300,349
R^2	0.73	0.73	0.78	0.78

Notes: This table provides the estimates of the difference-in-difference estimates of the impact of NIRP on capital structure using short-term debt ratio (SDR) following Equation 1 with Two-way Fixed Effects (TWFE), where DiD is $Treated_i \times POST_t$. *Clustered (firmid) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.*

Table 11: NIRP and Long-Term Debt TLTRO and Credit Growth – Full Sample

Model:	(1)	(2)	(3)	(4)
DiD	0.0050*	0.0045	0.0101***	0.0095***
	(0.0027)	(0.0028)	(0.0028)	(0.0028)
ECB_QE	0.0214***	0.0212***	0.0194***	0.0183***
	(0.0026)	(0.0027)	(0.0028)	(0.0028)
ECB_FG	-0.0059***	-0.0054***	-0.0054***	-0.0057***
	(0.0018)	(0.0018)	(0.0019)	(0.0020)
ECB_TLTRO	0.0110***	0.0106***	0.0139***	0.0135***
	(0.0013)	(0.0014)	(0.0014)	(0.0014)
AGE	-0.0010**	-0.0013**	-0.0009	-0.0013**
	(0.0005)	(0.0006)	(0.0006)	(0.0006)
PROFIT	-0.0328***	-0.0317***	-0.0542***	-0.0513***
	(0.0012)	(0.0012)	(0.0021)	(0.0021)
LIQ	0.0021	0.0043	-0.0072*	-0.0055
	(0.0030)	(0.0030)	(0.0043)	(0.0042)
SIZE	0.0149***	0.0147***	0.0264***	0.0263***
	(0.0005)	(0.0005)	(0.0009)	(0.0009)
AS	0.1200***	0.1180***	0.1486***	0.1466***
	(0.0032)	(0.0032)	(0.0045)	(0.0045)
INF	-0.0001	-0.0002*	0.0001	1.19×10^{-5}
	(0.0001)	(0.0001)	(0.0002)	(0.0002)
GDP growth	0.0009***	0.0007***	0.0011***	0.0009***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Bank Credit Growth(t-1)			0.0468***	0.0480***
			(0.0037)	(0.0038)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	No	Yes	No	Yes
No. of Firms	58,418	58,418	42,967	42,967
Observations	472,778	472,778	300,349	300,349
R^2	0.65	0.66	0.70	0.71

Notes: This table provides the estimates of the difference-in-difference estimates of the impact of NIRP on capital structure using long-term debt ratio (TDR) following Equation 1 with Two-way Fixed Effects (TWFE), where DiD is $Treated_i \times POST_t$. *Clustered (firmid) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.*

This pattern suggests that the maturity-extension effect of NIRP does not intensify during periods of strong credit growth. Instead, NIRP appears to *substitute* for weak credit supply—its influence on long-term financing is more pronounced when lending conditions are tight and firms face higher refinancing constraints.

Taken together, the results reveal a state-dependent but asymmetric form of transmission: NIRP amplifies the impact of credit supply on short-term borrowing (by reducing the incentive to dump short debt when supply is strong), while its effect on long-term debt is primarily counter-cyclical, supporting balance-sheet adjustment when credit growth is subdued.

5.4 Robustness: Sub-sample Analysis

In this section, we present the results of our sub-sample analysis. In this case, we provide analysis based on the firm size (Small *vs* Large), firm age (Young *vs* Old), firm leverage (High *vs* Low) and firm liquidity (High *vs* Low) defined relative to their respective median values[6]. The descriptive statistics of these sub-groups are provided in Tables C.1 and C.2 in Appenidx C. They generally show that large, older, and high-leverage treated firms have higher debt ratios than their counterparts. We proceed to discuss the results for the sub-samples.

5.4.1 The role of firm size

We define small firms as those with *SIZE* at or below the median and large firms as those above it. Regression results, reported in Tables D.1 to D.6 in Appendix D.1, confirm our prior findings: firms in NIRP environments reduce short-term debt while increasing long-term debt. Both small and large firms exhibit this maturity restructuring, but small firms increase their long-term debt proportionately more than they reduce their short-term debt. This suggests that, under NIRP, small firms, which are comparatively constrained in obtaining debt in normal conditions, disproportionately benefit from the abundant debt supply and low rates, leading to a significant positive *DiD* effect on their total debt ratio (TDR) in Table D.1. In contrast, large firms show no significant change in *TDR*, with an insignificant *DiD* effect in Table D.2, indicating stable overall leverage.

5.4.2 The role of firm age

The age of firms can influence their choice of debt composition. We, therefore, perform a sub-sample analysis comparing younger firms (those at or below the median *AGE*) with older firms

[6]We also provide robustness results using the lower and upper terciles of *SIZE*, *AGE*, *TDR* and *LIQ* for firm size, firm age, firm leverage and firm liquidity respectively. The results, which are available on request, are consistent with our main findings.

Table 12: NIRP and Total Debt with TLTRO and Credit Growth Interactions – Full Sample

Model:	(1)	(2)	(3)	(4)
DiD	0.0135***	0.0125***	0.0051	0.0058*
	(0.0030)	(0.0030)	(0.0033)	(0.0033)
ECB_QE	0.0028	0.0010	0.0074**	0.0047
	(0.0030)	(0.0031)	(0.0033)	(0.0033)
ECB_FG	-0.0173***	-0.0167***	-0.0163***	-0.0159***
	(0.0023)	(0.0023)	(0.0023)	(0.0023)
ECB_TLTRO			0.0078***	0.0062***
			(0.0017)	(0.0017)
AGE	0.0021**	0.0014	0.0021**	0.0014
	(0.0009)	(0.0009)	(0.0009)	(0.0009)
PROFIT	-0.1889***	-0.1867***	-0.1889***	-0.1867***
	(0.0037)	(0.0037)	(0.0037)	(0.0037)
LIQ	-0.4151***	-0.4134***	-0.4151***	-0.4134***
	(0.0066)	(0.0066)	(0.0066)	(0.0066)
SIZE	0.0185***	0.0185***	0.0185***	0.0185***
	(0.0012)	(0.0013)	(0.0012)	(0.0013)
AS	-0.1267***	-0.1289***	-0.1268***	-0.1290***
	(0.0062)	(0.0062)	(0.0062)	(0.0062)
INF	0.0016***	0.0013***	0.0016***	0.0014***
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
GDP growth	0.0018***	0.0015***	0.0018***	0.0015***
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
Bank Credit Growth(t-1)	-0.0336***	-0.0335***	-0.0337***	-0.0335***
	(0.0045)	(0.0046)	(0.0045)	(0.0046)
DiD × Bank Credit Growth(t-1)	-0.0076	-0.0222	0.0064	-0.0110
	(0.0306)	(0.0308)	(0.0319)	(0.0322)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	No	Yes	No	Yes
No. of Firms	42,967	42,967	42,967	42,967
Observations	300,349	300,349	300,349	300,349
R^2	0.78	0.78	0.78	0.78

Notes: This table provides the estimates of the difference-in-difference estimates of the impact of NIRP on capital structure using total debt ratio (TDR) following Equation 1 with Two-way Fixed Effects (TWFE), where DiD is $Treated_i \times POST_t$. *Clustered (firmid) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.*

Table 13: NIRP and Short-Term Debt with TLTRO and Credit Growth Interactions – Full Sample

Model:	(1)	(2)	(3)	(4)
DiD	-0.0109***	-0.0106***	-0.0062**	-0.0056**
	(0.0024)	(0.0024)	(0.0028)	(0.0028)
ECB_QE	-0.0049**	-0.0058**	-0.0075***	-0.0086***
	(0.0024)	(0.0025)	(0.0027)	(0.0027)
ECB_FG	-0.0065***	-0.0056***	-0.0071***	-0.0062***
	(0.0018)	(0.0018)	(0.0018)	(0.0018)
ECB_TLTRO			-0.0043***	-0.0047***
			(0.0015)	(0.0015)
AGE	0.0015***	0.0014**	0.0015***	0.0014**
	(0.0006)	(0.0006)	(0.0006)	(0.0006)
PROFIT	-0.1267***	-0.1278***	-0.1267***	-0.1278***
	(0.0028)	(0.0028)	(0.0028)	(0.0028)
LIQ	-0.4008***	-0.4007***	-0.4007***	-0.4006***
	(0.0054)	(0.0054)	(0.0054)	(0.0054)
SIZE	-0.0096***	-0.0093***	-0.0096***	-0.0093***
	(0.0009)	(0.0010)	(0.0009)	(0.0010)
AS	-0.3233***	-0.3220***	-0.3232***	-0.3220***
	(0.0053)	(0.0053)	(0.0053)	(0.0053)
INF	0.0017***	0.0017***	0.0017***	0.0016***
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
GDP growth	0.0001	0.0002*	0.0001	0.0002*
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Bank Credit Growth(t-1)	-0.0619***	-0.0631***	-0.0619***	-0.0631***
	(0.0040)	(0.0040)	(0.0040)	(0.0040)
DiD × Bank Credit Growth(t-1)	0.0579***	0.0576***	0.0501***	0.0492***
	(0.0176)	(0.0178)	(0.0182)	(0.0184)
Marginal Effect @ 1% Credit Growth	-1%	-1%	-0.5%	-0.5%
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	No	Yes	No	Yes
No. of Firms	42,967	42,967	42,967	42,967
Observations	300,349	300,349	300,349	300,349
R^2	0.78	0.78	0.78	0.78

Notes: This table provides the estimates of the difference-in-difference estimates of the impact of NIRP on capital structure using short-term debt ratio (SDR) following Equation 1 with Two-way Fixed Effects (TWFE), where DiD is $Treated_i \times POST_t$. *Clustered (firmid) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.*

Table 14: NIRP and Long-Term Debt with TLTRO and Credit Growth Interactions – Full Sample

Model:	(1)	(2)	(3)	(4)
DiD	0.0253***	0.0243***	0.0111***	0.0108***
	(0.0025)	(0.0025)	(0.0029)	(0.0029)
ECB_QE	0.0113***	0.0104***	0.0190***	0.0178***
	(0.0025)	(0.0026)	(0.0027)	(0.0028)
ECB_FG	-0.0079***	-0.0085***	-0.0063***	-0.0070***
	(0.0020)	(0.0020)	(0.0020)	(0.0020)
ECB_TLTRO			0.0131***	0.0125***
			(0.0015)	(0.0015)
AGE	-0.0010*	-0.0013**	-0.0009	-0.0013**
	(0.0006)	(0.0006)	(0.0006)	(0.0006)
PROFIT	-0.0542***	-0.0512***	-0.0542***	-0.0512***
	(0.0021)	(0.0021)	(0.0021)	(0.0021)
LIQ	-0.0071*	-0.0054	-0.0072*	-0.0055
	(0.0043)	(0.0042)	(0.0043)	(0.0042)
SIZE	0.0264***	0.0263***	0.0264***	0.0263***
	(0.0009)	(0.0009)	(0.0009)	(0.0009)
AS	0.1487***	0.1468***	0.1486***	0.1467***
	(0.0045)	(0.0045)	(0.0045)	(0.0045)
INF	0.0001	-4.72×10^{-6}	0.0001	1.24×10^{-5}
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
GDP growth	0.0011***	0.0008***	0.0011***	0.0009***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Bank Credit Growth(t-1)	0.0479***	0.0494***	0.0478***	0.0493***
	(0.0036)	(0.0037)	(0.0036)	(0.0037)
DiD × Bank Credit Growth(t-1)	-0.0570**	-0.0653**	-0.0334	-0.0428
	(0.0257)	(0.0259)	(0.0268)	(0.0269)
Marginal Effect @ 1% Credit Growth	2%	2%	—	—
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	No	Yes	No	Yes
No. of Firms	42,967	42,967	42,967	42,967
Observations	300,349	300,349	300,349	300,349
R^2	0.70	0.71	0.70	0.71

Notes: This table provides the estimates of the difference-in-difference estimates of the impact of NIRP on capital structure using long-term debt ratio (TDR) following Equation 1 with Two-way Fixed Effects (TWFE), where DiD is $Treated_i \times POST_t$. *Clustered (firmid) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.*

(those above the median AGE). The results are presented in Tables D.7 to D.12 in Appendix D.2. Consistent with our initial findings, firms tend to reduce short-term debt but increase long-term debt during periods of negative interest rates. This, however, does not seem to have a significant impact on the overall debt ratio. Moreover, we observe that older firms are more likely to undertake debt maturity restructuring. Older firms will generally have market experience and a reputation, and hence will be able to take on more long-term debt.

5.4.3 The role of firm leverage

We also provide robustness by comparing low-levered (at or below median TDR) with high-levered firms (above median TDR), shown in Tables D.13 to D.18 in Appendix D.3. Again, we find that both high and low-leveraged firms reduce their short-term debt to increase their long-term debt. However, high-leverage firms are more likely to undertake the debt maturity restructuring.

5.4.4 The role of firm liquidity

We also provide robustness checks for low- and high-liquid firms, where low-liquid firms are those at or below the median LIQ, and high-liquid firms are those above the median. The results are shown in Tables D.19 to D.24 in Appendix D.4. Again, we find consistent results that firms, both low- and high-liquid firms, reduce their short-term debt and increase their long-term debt in a negative interest rate environment, with no significant impact on the overall debt ratio. However, low-liquid firms seem to take more advantage of the maturity rebalancing than high-liquid firms.

5.5 Robustness: Sub-Sample Analysis with Credit Growth and TLTRO

We do robustness checks for sub-sample analysis where we control for credit growth and TLTRO with interactions between the DiD indicator and credit growth. The results are presented in Tables E.1 to E.24 in Appendix E.

Across all firm characteristics—size, age, leverage, and liquidity—a consistent asymmetry emerges that mirrors the aggregate results reported in Tables 12–14. For short-term debt, the interaction term ($DiD \times Bank\ Credit\ Growth_{t-1}$) is occasionally positive and statistically significant within certain subsamples. This pattern indicates that when credit conditions improve, firms' short-term borrowing becomes more responsive to NIRP. In these instances, NIRP and credit growth operate as *complementary* forces: abundant credit relaxes liquidity constraints,

allowing firms—especially those that are large, older, highly leveraged, or low in liquidity—to adjust short-term financing in a manner more sensitive to the negative-rate environment.

In contrast, the interaction terms for total debt and long-term debt are generally small and statistically insignificant across most subgroups, with the limited significant coefficients indicating a negative relationship. This reflects a *substitution* mechanism: the maturity-extension effect of NIRP is most pronounced when credit supply is tight, meaning that strong bank credit growth does not amplify the shift toward long-term borrowing. Instead, NIRP substitutes for weak credit conditions by encouraging firms—particularly large, highly leveraged or low-liquid firms—to secure long-term funding precisely when external financing is less plentiful. Overall, the heterogeneity results reinforce the central finding that NIRP's impact on debt maturity operates primarily through refinancing and balance-sheet frictions, with complementary effects concentrated in short-term debt and substitution effects dominating for long-term financing.

5.6 Robustness: Sub-Sample Analysis with OECD Countries

A potential concern with our baseline analysis is that the control group may include countries with fundamentally different economic structures, financial market development, or institutional quality from those of the Eurozone. To address this concern, we restrict our sample to OECD member countries, which share broadly similar characteristics in terms of economic development, financial sophistication, and regulatory frameworks.

The OECD sub-sample is particularly appropriate for several reasons. First, OECD countries have well-developed capital markets and banking systems, making them more comparable to Eurozone economies in terms of firms' access to alternative financing sources. Second, these countries experienced similar global macroeconomic conditions during the sample period, reducing concerns about confounding shocks that might have differentially affected the treatment and control groups. Third, the institutional and regulatory environments across OECD countries are more harmonised, mitigating concerns that differences in legal frameworks or creditor rights drive our results rather than the NIRP treatment itself.

Tables E.25, E.26, and E.27 in Appendix E.5 present the estimation results for long-term debt ratio, short-term debt ratio, and total debt ratio, respectively. The specifications include all ECB unconventional monetary policies (NIRP, QE, Forward Guidance, and TLTRO), firm-level controls, macroeconomic variables, and lagged bank credit growth.

The results strongly confirm our main findings. The DiD coefficient for long-term debt ratio remains positive and statistically significant across all specifications (Table E.25). This is consistent with our full-sample estimates, suggesting that the "borrow long" effect is robust when comparing ECB-treated firms to similar firms in other advanced economies. For the short-term

debt ratio, the DiD coefficient is small and statistically insignificant in almost all specifications (Table E.26), with the TLTRO policy absorbing most of this effect — having a negative effect on short-term debt.

For the total debt ratio, the DiD coefficient is positive and statistically significant (Table E.27). The effect strengthens when controlling for bank credit growth, consistent with the demand-driven interpretation that NIRP affects corporate leverage beyond aggregate credit supply. Notably, bank credit growth enters negatively and significantly, suggesting that after controlling for credit supply conditions, the direct effect of NIRP on firm leverage is even more pronounced.

Overall, restricting the analysis to OECD countries yields qualitatively identical conclusions to our baseline results. Firms in NIRP countries increase their long-term debt exposure, leading to higher overall leverage and a shift toward longer debt maturities. These findings reinforce our interpretation that NIRP induces corporate maturity rebalancing—firms "borrow long and dump short"—and this effect is not driven by differences between advanced and developing economies in the control group.

5.7 Event Study Evidence: Parallel Trends and Treatment Effects

To examine the validity of the parallel trends assumption and to provide event-study evidence of the treatment effects, we estimate two specifications that differ in their treatment of concurrent ECB unconventional monetary policies. The first specification is our baseline model with Quantitative Easing (QE) and Forward Guidance (FG), while the second additionally includes Targeted Longer-Term Refinancing Operations (TLTRO) and all other controls, with the full set of fixed effects.

Figure 3 presents the event study estimates across short-term debt ratio, long-term debt ratio, and total debt ratio. For the four-year pre-treatment window ($k = -4$ to $k = -2$), none of the pre-treatment coefficients are statistically significant at the 5% level for either the short-term, long-term or total debt ratio in specifications that include firm-level controls. This holds for both the specification excluding TLTRO and the specification including it. These patterns support the parallel trends assumption during the pre-NIRP period.

The top row of Figure 3 reports the event study estimates for the short-term debt ratio. The pre-treatment coefficients hover around zero and are statistically insignificant in the four-year window, consistent with parallel trends. Following the introduction of NIRP in 2014, the estimates turn sharply negative and remain persistently so through 2020. The effect is statistically significant, with point estimates ranging from approximately -0.008 at $k = 0$ to -0.013 at $k = 5$, indicating a sustained decline of roughly 0.8 to 1.3 percentage points in short-term debt ratios

for euro-area firms relative to their non-euro-area counterparts. Notably, the results are virtually identical across the two specifications, suggesting that the inclusion of TLTRO does not alter the estimated NIRP effect on short-term borrowing.

The middle row presents the corresponding estimates for the long-term debt ratio. Again, the pre-treatment coefficients in the four-year window are statistically insignificant, supporting the parallel trends assumption. In contrast to short-term debt, the post-treatment coefficients are positive, indicating that NIRP is associated with an increase in long-term borrowing by euro-area firms. The effect becomes statistically significant 2 years after the NIRP implementation and grows progressively larger, reaching approximately $+0.021$ at $k = 5$ and $+0.039$ at $k = 6$, corresponding to an increase of roughly 2.1 to 3.9 percentage points. The magnitude and trajectory of these estimates suggest that firms substituted away from short-term debt toward longer-maturity obligations in the negative interest rate environment. The pattern is consistent with a maturity-lengthening channel through which NIRP may have incentivised firms to lock in favourable long-term borrowing rates. As with short-term debt, the results are robust to the inclusion of TLTRO as a control.

The bottom row reports the estimates for the total debt ratio. The pre-treatment coefficients are generally insignificant. The post-treatment pattern is less pronounced than for the individual components. The effect at $k = 0$ is negative and significant (-0.007), but subsequent coefficients are largely statistically insignificant until after 5 years of policy implementation, when large, positive, and significant effects emerge. The muted post-treatment effects on total debt are consistent with the offsetting dynamics observed in the short-term and long-term components: the decline in short-term debt is partially offset by the increase in long-term debt, resulting in a smaller net effect on total leverage. The significant positive effects in the later periods ($k = 6$ and $k = 7$) likely reflect the dominance of the long-term debt channel in the later years of the sample. This may be because the policy was quite unconventional; firms would take time to consider whether it would likely remain in place over the medium term before deciding to change their capital structure.

5.8 Robustness: using other heterogeneous DiD method

Roth et al. (2023) suggest further robustness to the TWFE in our 2×2-DiD framework when the data is an unbalanced panel, such as our study. The authors suggest considering any of the "heterogeneity-robust" estimators for the staggered treatment timings. Hence, we employ the method by Callaway and Sant'Anna (2021) as a robustness check. The results confirm our earlier findings that firms in negative interest rate environments reduce their short-term debt while increasing their long-term debt to capitalise on the low interest rates.

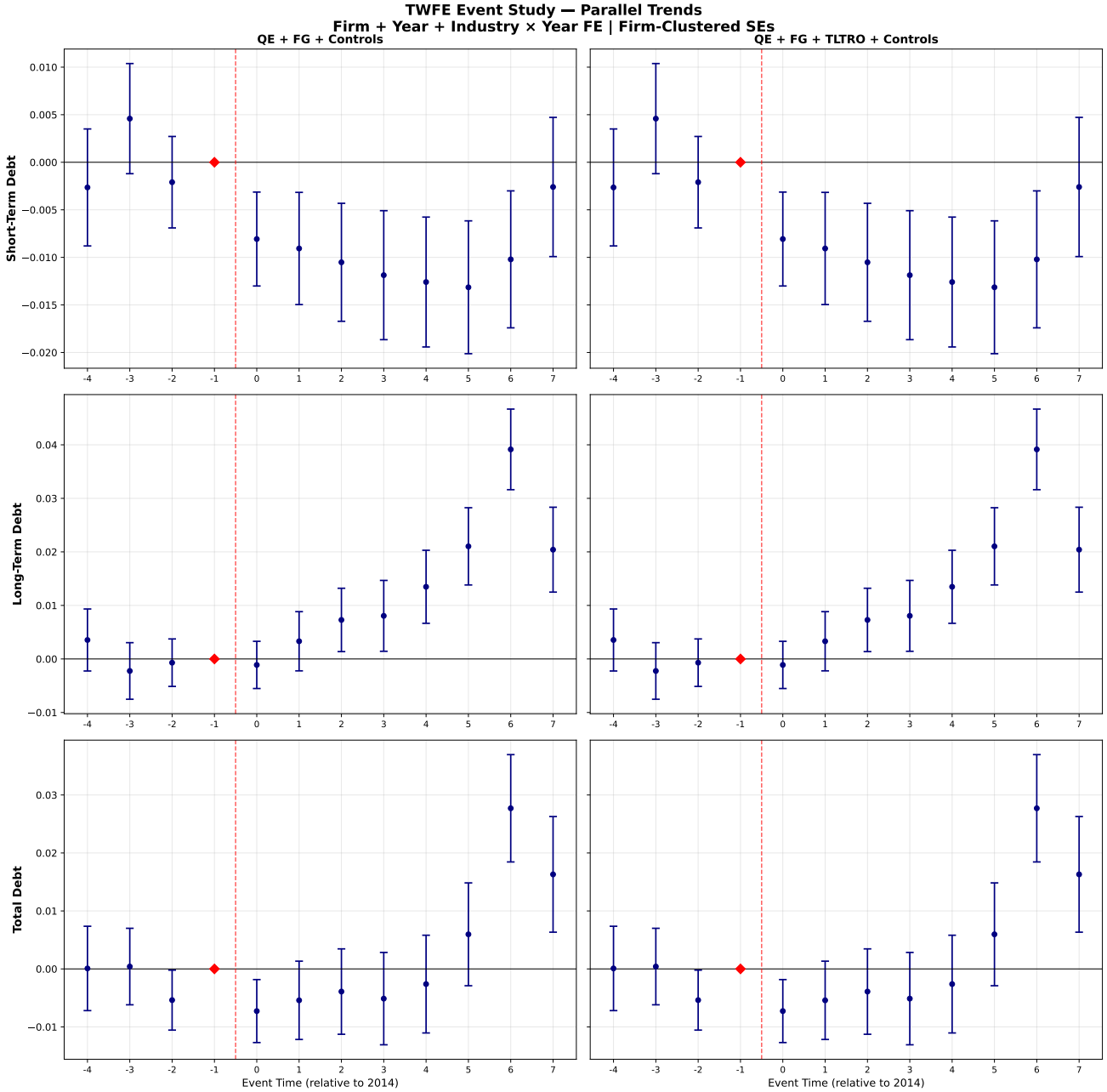

Figure 3: TWFE event study estimates for the effect of NIRP on firm debt ratios with 95% confidence intervals based on firm-clustered standard errors. The left column controls for QE and FG; the right column additionally controls for TLTRO. Rows correspond to short-term debt (top), long-term debt (middle), and total debt (bottom). The reference period is $k = -1$ (2013). The dashed red line marks the introduction of NIRP in 2014.

Table 15: Difference-in-Differences (ATT) estimates of impact of NIRP on Debt based on Callaway and Sant'Anna (2021).

	TDR	SDR	LDR
DiD	0.015***	-0.014***	0.024***
	(0.004)	(0.003)	(0.003)

*Note: Clustered (firm level) standard errors in parentheses. Other unconventional policy dummies are dropped in the specifications due to low observations in sub-groups. ***p<0.01.*

5.9 Robustness: using matching method

In order to provide further robustness, we use the panel data matching method of Imai et al. (2023)[7], which allows us to match each treated observation for a given unit in a particular period with untreated observations from other units in the same period that have a similar treatment and outcome histories. The matching process enhances the validity of the parallel trends assumption by mitigating key observable disparities between the treatment and control groups (Ben-Menachem and Morris, 2023). Thus, this method helps improve the validity of causal inference by reducing model dependencies (Imai et al., 2023; Ho et al., 2007). From Table 16, we again find that, while firms do not change their overall capital structure following the implementation of the NIRP, they reduce their short-term debt and increase their long-term debt, consistent with our earlier findings.

Table 16: Difference-in-Differences (ATT) estimates of impact of NIRP on Debt based on Imai et al. (2023).

	TDR	SDR	LDR
DiD	-0.006	-0.010***	0.007**
	(0.004)	(0.003)	(0.004)

*Note: Standard errors in parentheses. ***p<0.01, **p<0.05.*

[7]See Imai et al. (2023) for more discussion on how traditional difference-in-differences methods are improved when matching is used.

6 NIRP Transmission in a DSGE Framework

This section develops a dynamic stochastic general equilibrium (DSGE) model that provides microfoundations for the empirical evidence documented above. The central empirical finding—that firms shifted their debt composition toward longer maturities while total leverage remained initially stable—calls for a structural framework capable of distinguishing between the speed of maturity adjustment and the speed of leverage adjustment. The model jointly treats three features typically studied in isolation: (i) endogenous corporate debt maturity choice between short-term and long-term instruments, governed by firms' response to the term spread; (ii) financial intermediaries subject to balance sheet constraints and reserve requirements, whose net worth shapes credit supply; and (iii) a deposit rate floor that creates a wedge between policy rates and deposit rates under NIRP, generating the reserve-carry cost that is central to the delayed leverage response.

Our model builds on several strands of the DSGE literature. The New Keynesian core follows Smets and Wouters (2007), with Calvo-style nominal rigidities and external habit formation generating empirically plausible dynamics. The financial intermediation block draws on Gertler and Karadi (2011), who introduce balance-sheet constrained banks, giving rise to endogenous intermediation spreads. The maturity structure of corporate debt is informed by Greenwood et al. (2010) on firms' gap-filling behaviour and Badoer and James (2016) on firm-level maturity decisions. The NIRP transmission mechanism relates to Brunnermeier and Koby (2018), who characterise the "reversal rate" below which accommodation becomes contractionary.

6.1 Model Environment

The economy is populated by five types of agents that interact in goods, labour, financial, and bond markets. Time is discrete, indexed by $t = 0, 1, 2, \ldots$, and all decisions are made at quarterly frequency.

Households are infinitely lived and derive utility from consumption subject to external habit formation. They supply labour, hold bank deposits as their sole savings instrument, and receive dividends. Habit persistence generates hump-shaped consumption dynamics (Christiano et al., 2005).

Intermediate goods producers operate in a competitive market, combining capital and labour using a constant-returns-to-scale technology. They finance investment partly through retained earnings and partly through short- and long-term debt. Short-term debt matures within one period; long-term debt takes the form of a perpetuity (consol bond) with geometrically decaying coupons (Woodford, 2001).

Retailers purchase intermediate output at marginal cost and sell differentiated varieties to

final goods aggregators, facing Calvo-style price-setting frictions. A fraction θ_p of retailers cannot adjust prices each period, generating the New Keynesian Phillips curve.

Financial intermediaries (banks) accept deposits from households at the gross rate R_t^D and hold two categories of assets: corporate debt and central bank reserves. Banks face a moral hazard problem, limiting leverage (Gertler and Karadi, 2011), so that lending capacity depends on net worth. When the deposit rate is bounded at zero while the reserve rate is negative, banks incur a reserve-carry cost—the spread $(R_t^D - R_t^S)$ multiplied by reserve holdings—that acts as a persistent drag on profitability. This cost does not necessarily reduce net worth in absolute terms, since capital gains from yield compression can offset the margin squeeze. However, the reserve-carry cost attenuates the net worth improvement relative to conventional easing, constraining credit supply expansion.

The *monetary authority* sets the policy rate R_t^{Pol} via a Taylor-type rule with interest rate smoothing, subject to two constraints: the reserve rate R_t^S is bounded below by a NIRP floor $\underline{R}$, and the deposit rate R_t^D is bounded below at zero. The asymmetry $R_t^S < 1 \leq R_t^D$ generates the distinctive features of NIRP transmission.

Three financial instruments circulate. *Deposits* (D_t) are one-period claims earning $R_t^D \geq 1$. *Short-term corporate debt* (B_t^S) is one-period paper with gross yield R_t^S. *Long-term corporate debt* (B_t^L) takes the form of consol bonds with coupon decay $\kappa \in (0,1)$, priced at Q_t^L. Banks also hold *central bank reserves* (S_t) earning R_t^S. The maturity mix m_t—the share of long-term debt in total borrowing—is the key endogenous variable linking monetary policy to corporate financing.

6.2 Households

The representative household maximises expected lifetime utility over consumption C_t and labour L_t:

$$\max_{\{C_t, L_t, D_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(C_t - hC_{t-1})^{1-\sigma}}{1-\sigma} - \frac{\omega L_t^{1+\varphi}}{1+\varphi} \right], \tag{2}$$

where $\beta \in (0,1)$ is the discount factor, $\sigma > 0$ is relative risk aversion, $h \in (0,1)$ governs external habit persistence, $\omega > 0$ scales labour disutility, and $\varphi > 0$ is the inverse Frisch elasticity. The term hC_{t-1} introduces a reference level based on lagged consumption.

The household's budget constraint is:

$$P_t C_t + D_t \leq W_t L_t + R_{t-1}^D D_{t-1} + \Pi_t^{HH} - T_t, \tag{3}$$

where P_t is the price level, D_t denotes end-of-period deposits, W_t is the nominal wage, R_t^D is the gross deposit rate, Π_t^{HH} aggregates dividends from firms and banks, and T_t are lump-sum

taxes. Deposits are the household's sole savings instrument, consistent with the euro area's intermediation-based structure.

The first-order conditions yield the stochastic discount factor (SDF):

$$\Lambda_{t,t+1} \equiv \beta \left(\frac{C_{t+1} - hC_t}{C_t - hC_{t-1}} \right)^{-\sigma}, \tag{4}$$

and the Euler equation for optimal deposit choice:

$$1 = \mathbb{E}_t \left[\Lambda_{t,t+1} \frac{R_t^D}{\Pi_{t+1}} \right], \tag{5}$$

where $\Pi_{t+1} \equiv P_{t+1}/P_t$ is gross inflation. In the implemented equilibrium, R_t^D is replaced by the policy rate R_t^S; the economic rationale for this substitution is explained in the Remark below equation (6). Log-linearising and substituting $\hat{c}_t \approx \hat{y}_t$ yields the IS curve with habit persistence:

$$\hat{y}_t = \frac{1}{1+h} \mathbb{E}_t[\hat{y}_{t+1}] + \frac{h}{1+h} \hat{y}_{t-1} - \frac{1-h}{\sigma(1+h)} \left(\hat{r}_t^S - \mathbb{E}_t[\pi_{t+1}] - \hat{r}_t^{nat} \right), \tag{6}$$

where $\hat{y}_t$ is the output gap (log deviation from steady state), $\hat{r}_t^S$ is the reserve/policy rate in deviation, π_{t+1} is log inflation, and $\hat{r}_t^{nat} = -\chi_\theta \hat{\theta}_t$ is the natural rate driven by the exogenous demand shock $\hat{\theta}_t$ with sensitivity $\chi_\theta > 0$.

Remark 1 (Policy Rate in IS Curve) *While the household Euler equation (5) features the deposit rate R_t^D, the IS curve uses the reserve/policy rate R_t^S. This reflects that wholesale funding rates and money market rates—which determine the marginal cost of capital for firms and institutional investors—track the policy rate rather than the retail deposit rate. Under NIRP, the deposit rate is floored at zero, but wholesale rates (repos, interbank lending, commercial paper) remain responsive to policy. This is the relevant rate for aggregate demand dynamics, consistent with Gertler and Karadi (2011) and Chen et al. (2012).*

Habit formation introduces the lagged term $\hat{y}_{t-1}$ and attenuates interest rate sensitivity by $(1-h)/[\sigma(1+h)]$. For $h = 0.70$ and $\sigma = 1.38$, the effective semi-elasticity is approximately 0.128, substantially lower than in a model without habits.

6.3 Production Sector

6.3.1 Intermediate Goods Producers

A continuum of competitive intermediate goods producers combine capital K_t and labour L_t using Cobb–Douglas technology:

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}, \tag{7}$$

where A_t is total factor productivity following $\ln A_t = \rho_a \ln A_{t-1} + \varepsilon_t^a$ with persistence ρ_a and innovation ε_t^a, and $\alpha \in (0,1)$ is the capital share. The capital stock evolves according to:

$$K_{t+1} = (1-\delta)K_t + \left[1 - \frac{\kappa_I}{2}\left(\frac{I_t}{I_{t-1}} - 1\right)^2\right] I_t, \tag{8}$$

where $\delta \in (0,1)$ is the depreciation rate, I_t is gross investment, and $\kappa_I \geq 0$ governs investment adjustment costs. This specification penalises changes in the *growth rate* of investment (rather than its level), introduced by Christiano et al. (2005).

6.3.2 Retailers and Price Setting

Monopolistically competitive retailers purchase intermediate output and sell differentiated varieties. The final good is produced by a CES aggregator with elasticity $\varepsilon_p > 1$, yielding standard Dixit–Stiglitz demand. Retailers face Calvo pricing: each period, a fraction $1-\theta_p$ re-optimises while θ_p maintains previous prices. Aggregating yields the New Keynesian Phillips curve:

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa_p \widehat{mc}_t, \tag{9}$$

where $\pi_t \equiv \ln \Pi_t$ is log inflation, $\widehat{mc}_t$ is real marginal cost in deviation from steady state, and $\kappa_p \equiv (1-\theta_p)(1-\beta\theta_p)/\theta_p$. With $\theta_p = 0.75$ (average price duration of four quarters) and $\beta = 0.995$, this yields $\kappa_p \approx 0.085$. Since the model includes capital and labour, marginal cost is $\widehat{mc}_t = \hat{w}_t - (\hat{y}_t - \hat{l}_t)$, where $\hat{w}_t$ is the real wage and $\hat{l}_t$ is labour.

6.4 Corporate Finance with Endogenous Debt Maturity

The treatment of corporate debt is the central innovation of our model. Firms choose both the total volume of debt and its maturity composition, a distinction that is critical for reconciling the empirical evidence that maturity responds to NIRP faster than leverage.

6.4.1 Debt Instruments

Firms issue two classes of debt, both of which are intermediated by banks.

Short-term debt. Let B_t^S denote one-period bonds issued at price Q_t^S, with gross borrowing rate $R_t^S = 1/Q_t^S$. In the implemented baseline (`nirp_regime` $= 1$), the retail deposit rate is fixed at its floor ($R_t^D = 1$), while the wholesale short rate R_t^S follows policy subject to the occasionally binding NIRP floor.

Long-term debt. Following Woodford (2001), long-term debt is a consol with geometrically decaying coupons: a unit issued at t pays κ^{j-1} at each $t+j$, where $\kappa \in (0,1)$ governs duration. The Macaulay duration is $1/(1-\beta\kappa)$; with $\kappa = 0.975$ and $\beta = 0.995$, this is approximately 33 quarters (≈ 8 years). The market price is Q_t^L, and the one-period holding-period return is:

$$R_{t+1}^L = \frac{1 + \kappa Q_{t+1}^L}{Q_t^L}. \tag{10}$$

The recursive structure reduces the infinite-maturity bond to a two-period pricing problem.

6.4.2 Debt Maturity Choice

We define the *long-debt share* (maturity mix) as:

$$m_t \equiv \frac{Q_t^L B_t^L}{Q_t^S B_t^S + Q_t^L B_t^L}, \tag{11}$$

and total real debt as:

$$b_t \equiv \frac{Q_t^S B_t^S + Q_t^L B_t^L}{P_t}. \tag{12}$$

Firms face quadratic adjustment costs when changing maturity:

$$\Phi_t^m = \frac{\phi_m}{2}(m_t - m_{t-1})^2, \tag{13}$$

and when changing total leverage:

$$\Phi_t^b = \frac{\phi_b}{2}(b_t - b_{t-1})^2. \tag{14}$$

These adjustment costs represent frictions in restructuring debt composition (ϕ_m: transaction fees, covenant modifications, market impact) and changing overall indebtedness (ϕ_b: renegotiation with creditors, credit line adjustments, rating agency targets). The firm's first-order conditions imply that the persistence parameters are structurally determined: $\rho_m = \phi_m/(1+\phi_m)$ and $\rho_b = \phi_b/(1+\phi_b)$. With $\phi_m = 2.0$ and $\phi_b = 8.0$, this yields $\rho_m = 0.667$ and $\rho_b = 0.889$, respectively. The substantial difference ($\rho_b > \rho_m$) ensures that firms find it easier to shift debt *composition*—for instance, rolling commercial paper into term loans—than to change its *level*.

This asymmetry is well-supported: Badoer and James (2016) document active maturity management in response to yield curve movements, while Leary and Roberts (2005) emphasise the sluggishness of leverage adjustment.

6.4.3 Optimal Maturity Condition

The firm minimises financing costs plus adjustment costs subject to production and capital accumulation constraints. The implemented log-linearised first-order condition is:

$$\hat{m}_t = \rho_m \hat{m}_{t-1} - (1 - \rho_m)\eta_m(\hat{r}_t^L - \bar{r}^L) + \varepsilon_t^m, \tag{15}$$

where $\hat{m}_t$ is the log deviation of the long-debt share, $\rho_m = \phi_m/(1 + \phi_m)$ is persistence derived from the adjustment cost, $\eta_m > 0$ captures sensitivity to the long-rate term, and ε_t^m is an exogenous maturity shock capturing regulatory changes, market microstructure shifts, or sudden shifts in duration demand. This reduced-form expression is the one coded in Dynare and is equivalent to a term-spread condition around the within-NIRP benchmark where the short-rate reference is absorbed into constants. The structural derivation ensures proper identification: higher adjustment costs ϕ_m simultaneously slow maturity adjustment (higher ρ_m) and reduce the contemporaneous impact (lower $1-\rho_m$). With $\phi_m = 2.0$ (implying $\rho_m = 0.667$) and $\eta_m = 2.0$, the model generates a rapid maturity response reaching 90 percent of its peak within two quarters.

6.4.4 Total Debt Dynamics

Total real debt follows a partial adjustment with substantially higher persistence:

$$\hat{b}_t = \rho_b \hat{b}_{t-1} + (1 - \rho_b)\left[-\eta_b(\hat{r}_t^L - \bar{r}^L) + \zeta_b(\hat{n}_t - \bar{n})\right], \tag{16}$$

where $\hat{b}_t$ is the log deviation of total debt, $\rho_b = \phi_b/(1 + \phi_b)$ is persistence derived from the leverage adjustment cost, $\eta_b > 0$ governs sensitivity to the long yield, and $\zeta_b > 0$ captures the credit supply channel through which bank net worth $\hat{n}_t$ affects borrowing capacity. With $\phi_b = 8.0$, the implied persistence $\rho_b = 0.889$ is substantially higher than $\rho_m = 0.667$, reflecting frictions in leverage adjustment including covenant constraints, refinancing costs, and infrequent capital structure rebalancing. The credit supply term creates an important link between banking sector health and corporate borrowing: when the reserve-carry cost attenuates net worth, credit supply contracts even if funding costs have declined.

6.4.5 Investment Dynamics

Investment is governed by a Tobin's Q framework with adjustment costs. Capital accumulates according to (8), where the adjustment cost parameter $\kappa_I = 2.48$ penalises changes in the investment growth rate. The investment Euler equation yields:

$$\hat{q}_t^k = \beta(1-\delta)\mathbb{E}_t[\hat{q}_{t+1}^k] + (1-\beta(1-\delta))\mathbb{E}_t[\hat{r}_{t+1}^k] + \kappa_I\beta(\mathbb{E}_t[\hat{i}_{t+1}] - \hat{i}_t) - \kappa_I(\hat{i}_t - \hat{i}_{t-1}), \tag{17}$$

where $\hat{q}_t^k$ is Tobin's Q (shadow value of capital), $\hat{r}_t^k$ is the rental rate, and the κ_I terms capture the forward-looking and backward-looking components of adjustment costs. Investment responds to Tobin's Q:

$$\hat{i}_t = \hat{i}_{t-1} + \frac{1}{\kappa_I}\hat{q}_t^k. \tag{18}$$

The long-term interest rate affects investment through Tobin's Q: lower rates raise expected capital returns and asset values, increasing $\hat{q}_t^k$ and stimulating investment. The adjustment cost specification generates the characteristic hump-shaped investment response observed in VAR evidence (Christiano et al., 2005).

6.5 Financial Intermediaries

Financial intermediaries channel household savings to corporate borrowers and are the locus through which NIRP's distinctive features affect the real economy. Our treatment follows Gertler and Karadi (2011), augmented with reserve requirements and ECB institutional features.

6.5.1 Balance Sheet

Each bank operates with a balance sheet identity:

$$Q_t^S B_t^S + Q_t^L B_t^L + S_t = D_t + N_t, \tag{19}$$

where assets include short-term corporate loans ($Q_t^S B_t^S$), long-term bonds ($Q_t^L B_t^L$), and central bank reserves (S_t), while liabilities comprise deposits (D_t) and net worth (N_t)—the bank's equity buffer and key state variable governing credit supply.

6.5.2 Reserve Requirement

Banks hold reserves in proportion to deposits:

$$S_t \geq \nu D_t, \quad \nu > 0, \tag{20}$$

where ν is the reserve ratio. Under NIRP, reserves earn the negative rate $R_t^S < 1$ while deposits cost $R_t^D = 1$, so the spread $(R_t^D - R_t^S) > 0$ imposes a cost proportional to reserves. The aggregate reserve-to-GDP ratio is $\bar{S}/\bar{Y} = 0.21$, reflecting ECB balance sheet expansion during the NIRP period.

6.5.3 Leverage Constraint

Moral hazard limits bank leverage. The bank can divert assets for private benefit but would lose its franchise value; depositors anticipate this and limit funds. The incentive compatibility constraint implies:

$$Q_t^S B_t^S + Q_t^L B_t^L \leq \phi N_t, \tag{21}$$

where $\phi = 4.0$ is the maximum leverage ratio. When binding, credit expansion is governed by the rate at which net worth accumulates. The log-linearised form is:

$$\hat{lev}_t = \bar{\phi} + \phi_{lev}(\hat{n}_t - \bar{n}), \tag{22}$$

where $\phi_{lev} > 0$ governs leverage response to net worth deviations.

6.5.4 Net Worth Dynamics

Bank net worth evolves according to:

$$N_t = \rho_N N_{t-1} + (1 - \rho_N)\left[\chi_Q(Q_t^L - \bar{Q}^L) - \chi_\tau(R_{t-1}^D - R_{t-1}^S)\bar{S}\right] + \varepsilon_t^N, \tag{23}$$

where $\rho_N \in (0, 1)$ captures persistence (reflecting dividend payouts and retained earnings), and ε_t^N is an exogenous net worth shock capturing loan losses, regulatory capital changes, or mark-to-market events (cf. the "capital quality shock" in Gertler and Karadi, 2011). The reserve-carry cost uses *lagged* rates $(R_{t-1}^D - R_{t-1}^S)$, reflecting that banks observe the margin squeeze from the previous period's operations before the current period's net worth is determined. Two terms govern gains and losses:

- *Bond valuation gains* (χ_Q term): Capital gains from fluctuations in consol prices Q_t^L relative to $\bar{Q}^L$ augment net worth. The parameter $\chi_Q > 0$ depends on the share of long-term assets in the bank's portfolio.

- *Reserve-carry tax* (χ_τ term): When $R_t^D > R_t^S$, banks incur a cost proportional to reserve holdings $\bar{S}$. With $R^D = 1$ and $R^S = 0.999$ ($\approx -0.4\%$ annualised), the steady-state cost is $(1 - 0.999) \times 0.21 \approx 0.021\%$ of GDP per quarter.

Remark 2 (Reserve-Carry Mechanism: Two Competing Forces) *Under conventional easing, a rate cut reduces both policy and deposit rates, leaving margins unchanged. Under NIRP, the deposit floor creates an asymmetry: the reserve rate falls but the deposit rate cannot follow,* widening $(R_t^D - R_t^S)$ *and increasing the carry cost. However, the rate cut simultaneously generates capital gains on bank bond portfolios through the signalling channel. Lopez et al. (2020) find that capital gains substantially offset margin compression in the short run, while Heider et al. (2019) and Brunnermeier and Koby (2018) emphasise that the margin squeeze dominates over longer horizons.*

6.5.5 Intermediation Spreads

When the leverage constraint binds, lending rates include a spread:

$$\mu_t = \exp(-\gamma_N \hat{n}_t) - 1, \tag{24}$$

where $\hat{n}_t$ is net worth in deviation from steady state and $\gamma_N = 8.0$ governs curvature. Spreads are zero at steady state, rise exponentially as net worth falls, and decline modestly as it rises, capturing the amplification mechanism documented by Gertler and Karadi (2011).

6.6 Monetary Policy

6.6.1 Policy Rule

The monetary authority sets the policy rate via a Taylor rule with smoothing:

$$\ln R_t^{Pol} = \rho_r \ln R_{t-1}^{Pol} + (1 - \rho_r) \left[\ln \bar{R} + \phi_\pi \pi_t + \phi_y \hat{y}_t \right] + \varepsilon_t^r, \tag{25}$$

where R_t^{Pol} is the gross policy rate, $\bar{R}$ is its steady-state value, $\rho_r = 0.80$ captures gradualism, $\phi_\pi = 1.25$ is the inflation response (following Campos et al. (2024); satisfying the Taylor principle), $\phi_y = 0.125$ is the output response (quarterly; annualised $4 \times 0.125 = 0.50$, following Darracq Pariès et al. (2023)), and ε_t^r is a monetary policy shock.

6.6.2 NIRP Floor and Deposit Floor

The reserve rate is bounded by the NIRP floor:

$$R_t^S = \max\{\underline{R}, R_t^{Pol}\}, \tag{26}$$

with $\underline{R} = 0.99749$ (-1.0% annualised), calibrated to the reversal rate estimate in Brunnermeier and Koby (2018). In the Dynare implementation, this max operator is solved with OccBin via regime-specific equations and the constraint-switching rule. The retail deposit rate is implemented as:

$$R_t^D = \xi R^{D,\text{floor}} + (1-\xi) R_t^S, \qquad \xi \in \{0,1\}, \tag{27}$$

with baseline within-NIRP calibration $\xi = 1$ and $R^{D,\text{floor}} = 1$, reflecting banks' reluctance to charge negative retail rates during 2014–2022. Under NIRP, this setup generates the reserve-carry cost:

$$\text{rcost}_t = (R_t^D - R_t^S) \cdot \frac{\bar{S}}{\bar{Y}}, \tag{28}$$

where $\bar{S}/\bar{Y}$ is the reserve-to-GDP ratio. This cost is the wedge distinguishing NIRP from conventional easing.

6.6.3 Long-Term Bond Pricing

The consol price satisfies a no-arbitrage condition discounted at the policy rate:

$$Q_t^L = \frac{\beta(1 + \kappa \mathbb{E}_t[Q_{t+1}^L])}{R_t^S \cdot (1+\mu_t) \cdot (1 + \mathbb{E}_t[\pi_{t+1}])}, \tag{29}$$

where R_t^S is the reserve/policy rate, κ is coupon decay, and μ_t is the intermediation spread from (24). The key feature is that bond prices are discounted at the policy rate R_t^S, not the deposit rate R_t^D. This captures the *signalling channel*: a rate cut (lower R_t^S) directly raises Q_t^L, compressing the long yield. For institutional investors and wholesale markets, the relevant opportunity cost of capital tracks the policy rate, not the retail deposit rate floored at zero. The recursive structure captures forward-looking bond valuation: a rate cut signalling prolonged accommodation raises expected future prices. The real long yield is:

$$r_t^L \equiv \ln\left(\frac{1 + \kappa Q_t^L}{Q_{t-1}^L}\right) - \mathbb{E}_t[\pi_{t+1}]. \tag{30}$$

6.7 Market Clearing and Equilibrium

The aggregate resource constraint is:

$$Y_t = C_t + I_t + G_t + \Phi_t^m + \Phi_t^b, \tag{31}$$

where G_t is government spending (constant GDP share) and $\Phi_t^m + \Phi_t^b$ are adjustment costs. In the log-linearised model, adjustment costs are second-order and drop out:

$$\hat{y}_t = c_y \hat{c}_t + i_y \hat{i}_t, \tag{32}$$

with $c_y = 0.60$ the steady-state consumption share and $i_y = 0.20$ the investment share. Government spending is the residual: $G/Y = 1 - c_y - i_y = 0.20$. Labour market clearing requires:

$$\omega L_t^{\varphi} (C_t - hC_{t-1})^{\sigma} = \frac{W_t}{P_t}. \tag{33}$$

Definition 1 (Competitive Equilibrium) *A competitive equilibrium consists of sequences of prices* $\{P_t, W_t, Q_t^S, Q_t^L, R_t^S, R_t^D\}$*, quantities* $\{C_t, L_t, I_t, Y_t, K_t, D_t, B_t^S, B_t^L, S_t, N_t\}$*, and auxiliary variables* $\{m_t, b_t, \mu_t, r_t^L\}$ *such that:*

(i) Households maximise (2) subject to (3), taking prices as given.

(ii) Producers choose labour, capital, and debt structure subject to (7), (8), and (13)–(14).

(iii) Retailers set prices subject to Calvo frictions.

(iv) Banks maximise value subject to (19), (20), and (21).

(v) The monetary authority follows (25) subject to (26) (implemented with OccBin) and (27).

(vi) All markets clear: goods (31), labour (33), and deposit/bond markets.

The complete log-linearised system, comprising 30 equations in 30 endogenous variables, is presented in Appendix G. In the baseline analysis, we set $\sigma_m = \sigma_N = \sigma_a = 0$ and focus on the transmission of monetary policy and demand shocks.

6.8 Model Mechanisms and Predictions

Before presenting quantitative results, we describe the transmission channels and state the model's central predictions as testable propositions.

6.8.1 NIRP Transmission Channels

The model features three channels through which a negative policy rate shock propagates to corporate debt structure.

Signalling Channel. A rate cut below zero signals prolonged accommodation. Through the recursive consol pricing equation (29), this signal raises bond prices and lowers long yields. When the NIRP floor binds, agents anticipate the rate will remain at the floor for an extended period, amplifying the forward guidance effect and generating immediate capital gains on long-term bond portfolios. The decline in long-term borrowing costs reduces the cost of capital, encourages investment, and makes long-term issuance more attractive. This channel is reinforced under NIRP because the short-rate side is constrained by the policy floor regime, while retail deposits remain floored in the baseline calibration.

Reserve-Carry Channel. When $R_t^S < 1 = R_t^D$, a further cut widens the spread $(R_t^D - R_t^S)$, increasing the reserve-carry cost. This cost enters (23) as a drag that partially offsets bond valuation gains. In our calibration with a -1% floor, a 25bp shock that triggers the floor generates a forward guidance effect: agents anticipate the rate will remain at the floor, raising bond prices immediately. Net worth rises on impact as capital gains on long-term bond portfolios dominate the reserve-carry cost. The mechanism is not that NIRP destroys profitability, but that the binding floor activates a forward guidance channel that conventional (non-binding) cuts lack.

Portfolio Rebalancing. The compression of long yields relative to the within-NIRP short-rate benchmark makes long-term debt cheaper. Firms respond through (15) by substituting toward long-term issuance, while total debt adjusts more slowly due to higher adjustment costs and the credit supply constraint. This generates the "borrow long, dump short" pattern.

6.8.2 Empirical Predictions

Proposition 1 (Maturity Rebalancing) *Under NIRP, the long-debt share m_t increases on impact. In the implemented maturity law (15), a decline in the real long yield $(\hat{r}_t^L - \bar{r}^L)$ directly raises $\hat{m}_t$.*

Proposition 2 (Delayed Leverage Response) *Total leverage b_t responds with a lag relative to maturity, despite net worth rising on impact when the floor binds. This occurs because: (i) high persistence $\rho_b = 0.889$ (from adjustment cost $\phi_b = 8.0$) reflects real-world frictions in leverage adjustment, substantially exceeding maturity persistence $\rho_m = 0.667$; and (ii) while the credit supply channel receives a positive impulse from rising net worth, the overall leverage response is dampened by the high adjustment costs. The differential timing—maturity adjusts rapidly while leverage adjusts slowly—is the model's central prediction.*

Proposition 3 (Central Bank Balance Sheet Amplification) *The delayed leverage response is more pronounced when $\bar{S}/\bar{Y}$ is large, because the reserve-carry tax is proportional to the stock*

of reserves.

These predictions align with the empirical findings documented in Section 5: firms immediately shift toward long-term debt, but total debt expansion materialises only after approximately five years of *persistent* negative rates in the data. The model's differential persistence ($\rho_m = 0.667$ versus $\rho_b = 0.889$) rationalises this pattern: a transitory shock produces rapid maturity rebalancing (90% of peak within 2 quarters) while leverage adjusts more slowly (90% within 3 quarters). Only sustained NIRP—as observed empirically—generates the cumulative stimulus required for significant leverage expansion.

6.9 Model Results and Interpretation

Figure 4 displays impulse responses to a one-standard-deviation negative policy rate shock—equivalent to approximately 100 basis points annualised. The reserve rate R^s falls on impact and dissipates gradually ($\rho_r = 0.80$). Additional figures comparing policy and demand shock responses, as well as term structure decomposition, are available in the supplementary materials.

Real economy effects. Output rises by approximately 0.10% on impact, peaking at 0.12% in quarter 2, driven by consumption and investment. The investment response exhibits a characteristic hump-shaped pattern, induced by investment adjustment costs ($\kappa_I = 2.48$), rising from 0.014% in quarter 1 to a peak of 0.031% in quarter 5 before gradually decaying. Inflation increases modestly, reflecting the Phillips curve relationship. These responses confirm that NIRP operates as an expansionary tool even with the deposit rate at its floor.

Term structure compression. The real long yield r^L falls on impact, reflecting the forward-looking nature of consol pricing. Since bond prices are discounted at the policy rate R^S (not the floored deposit rate), a rate cut directly raises bond prices through the discount factor, capturing the signalling channel. This compression is the key driver of both maturity rebalancing and eventual debt expansion.

When the NIRP floor binds, a forward guidance effect amplifies bond price movements. Agents anticipate the rate will remain at the floor, capitalising on this expectation by immediately pushing bond prices higher. The consol price Q^L rises on impact as the forward guidance channel dominates inflation and spread effects. This mechanism explains why net worth rises when the floor binds: banks holding long-term bonds experience immediate capital gains that more than offset the reserve-carry cost.

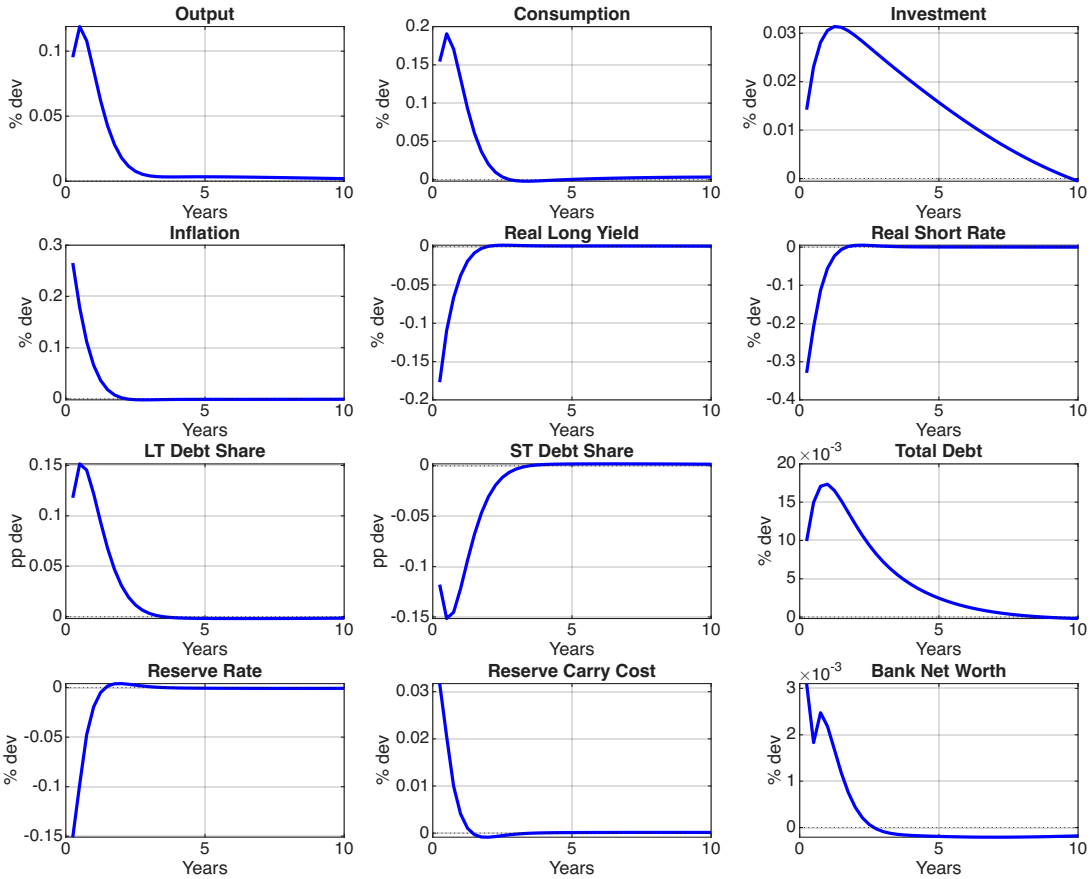

Figure 4: Impulse responses to a 25bp NIRP rate cut with -1% floor. The shock triggers the floor for one quarter, generating forward guidance effects that raise bond prices and bank net worth on impact.

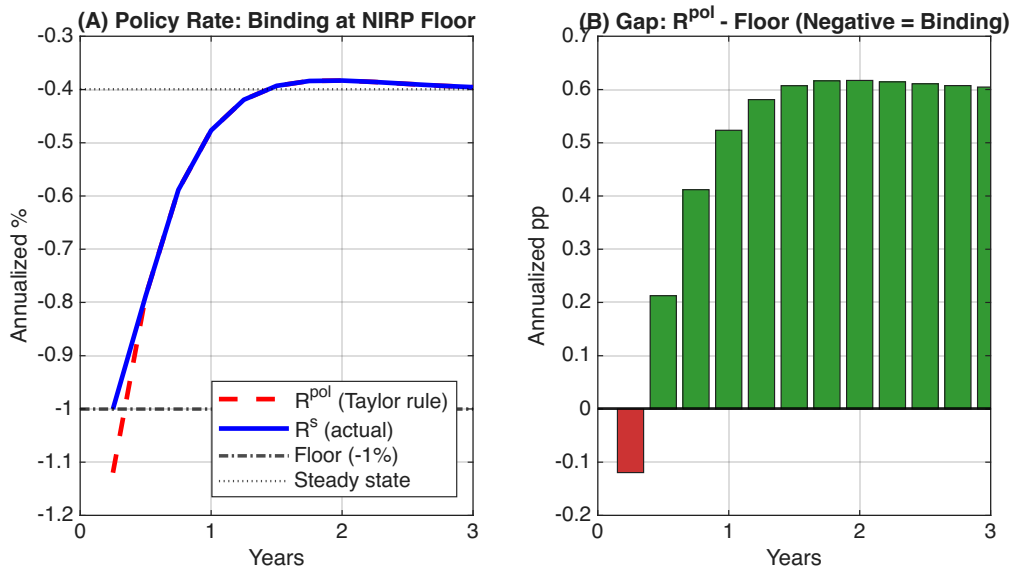

Figure 5: NIRP floor binding mechanism. Panel (A): The Taylor rule (dashed red) prescribes a rate below the -1% floor in quarter 1; the actual rate (solid blue) is truncated at the floor. Panel (B): The gap between the Taylor rule prescription and the floor; negative values (red bar) indicate binding periods. With a 25bp shock and -1% floor, the constraint binds for exactly one quarter.

6.10 Maturity Rebalancing: The Central Prediction

Panel (A) of Figure 6 shows the long-term debt share rising by approximately 0.12 percentage points on impact, peaking at approximately 0.15 percentage points in quarter 2, while the short-term share falls symmetrically—the "borrow long, dump short" pattern mirroring our empirical evidence. The mechanism is direct: yield compression reduces the relative cost of long-maturity financing, so the maturity equation implies that the positive term spread raises $\hat{m}_t$. With $\eta_m = 2.0$ and $\rho_m = 0.667$ (derived from $\phi_m = 2.0$), the adjustment is rapid. Figure 7 confirms that maturity reaches 90% of its peak within 2 quarters.

6.11 Delayed Leverage Response

Panel (B) of Figure 6 contrasts maturity with total debt dynamics. While the long-term share peaks within the first year, total debt builds gradually, peaking at approximately 0.017% in quarter 4 and remaining elevated thereafter.

High persistence in debt accumulation. The total debt equation (16) features $\rho_b = 0.889$ (derived from $\phi_b = 8.0$), implying firms adjust leverage only gradually even when financing improves. This reflects covenant constraints, refinancing costs, and infrequent capital structure

rebalancing.

Credit supply channel via bank net worth. The $\zeta_b(\hat{n}_t - \bar{n})$ term captures credit supply. When the NIRP floor binds, net worth rises on impact (+0.0031%) due to the forward guidance channel: agents anticipate the rate will remain at the floor, generating immediate capital gains on bank bond portfolios. Despite this positive impulse to credit supply, total debt still adjusts slowly because (i) the leverage adjustment cost is high ($\phi_b = 8.0$, implying $\rho_b = 0.889$), and (ii) net worth adjusts gradually ($\rho_n = 0.85$), so the credit supply stimulus builds slowly over time.

Figure 7 shows maturity reaching 90% of its peak in 2 quarters while total debt does so in 3 quarters. Total debt remains elevated at the 5-year horizon, matching our empirical finding that leverage rises significantly only after prolonged exposure to negative rates.

6.12 The Reserve-Carry Mechanism: Two Competing Forces

Panel (C) of Figure 6 illustrates the two competing forces operating on bank net worth under the baseline (binding floor) scenario. The rate cut widens ($R^D - R^S$), increasing the reserve-carry cost (dashed red line), while simultaneously generating bond valuation gains that raise net worth (solid blue line). The net effect on net worth is governed by:

$$\hat{n}_t = \rho_n \hat{n}_{t-1} + (1 - \rho_n) \left[\underbrace{\chi_Q(\hat{Q}_t^L - \bar{Q}^L)}_{\text{bond valuation gain (+)}} - \underbrace{\chi_\tau(\text{rcost}_{t-1} - \overline{\text{rcost}})}_{\text{reserve-carry drag (-)}} \right]. \tag{34}$$

When the NIRP floor binds, the forward guidance channel dominates: agents anticipate the rate will remain at the floor, which immediately raises bond prices. Net worth rises on impact (+0.0031%) as capital gains on long-term bond portfolios exceed the reserve-carry cost. This is consistent with Lopez et al. (2020), who find that capital gains dominated margin compression in the short run. The positive net worth impulse supports credit supply, though the high persistence of leverage adjustment ($\rho_b = 0.889$) still produces a gradual response of total debt.

Abadi et al. (2023) estimate the reversal rate at $\approx -1\%$, well below the ECB's -0.50%, confirming NIRP remained in the expansionary region. Ulate (2021) finds NIRP retains 60–90% of conventional effectiveness, the gap reflecting precisely the reserve-carry attenuation. The model illuminates why NIRP produces "borrow long, dump short" without immediate leverage expansion.

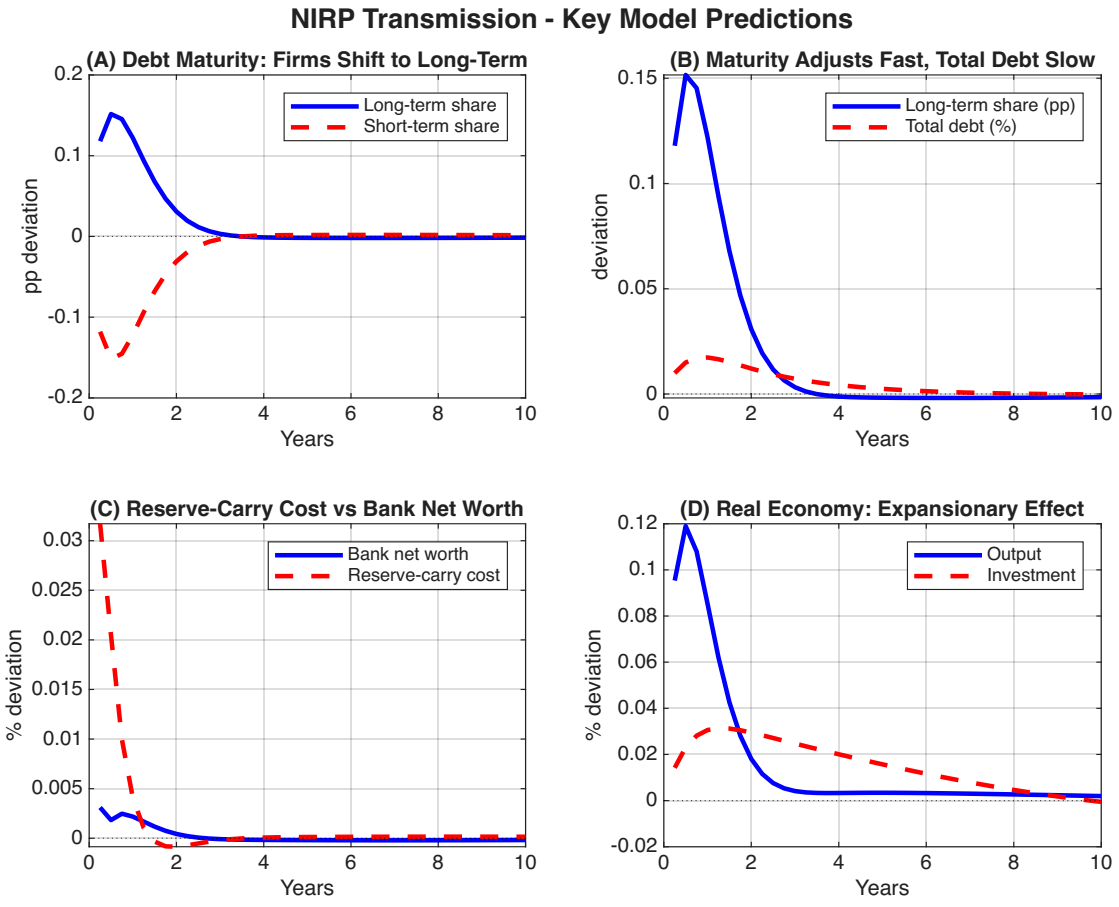

Figure 6: Key model predictions. Panel (A): Firms shift from short-term to long-term debt on impact—the "borrow long, dump short" pattern. Panel (B): Maturity adjusts immediately while total debt responds gradually. Panel (C): The rate cut increases reserve-carry cost, but net worth rises on impact (+0.0031%) due to the forward guidance channel when the floor binds. Panel (D): Output and investment rise, confirming expansionary effects.

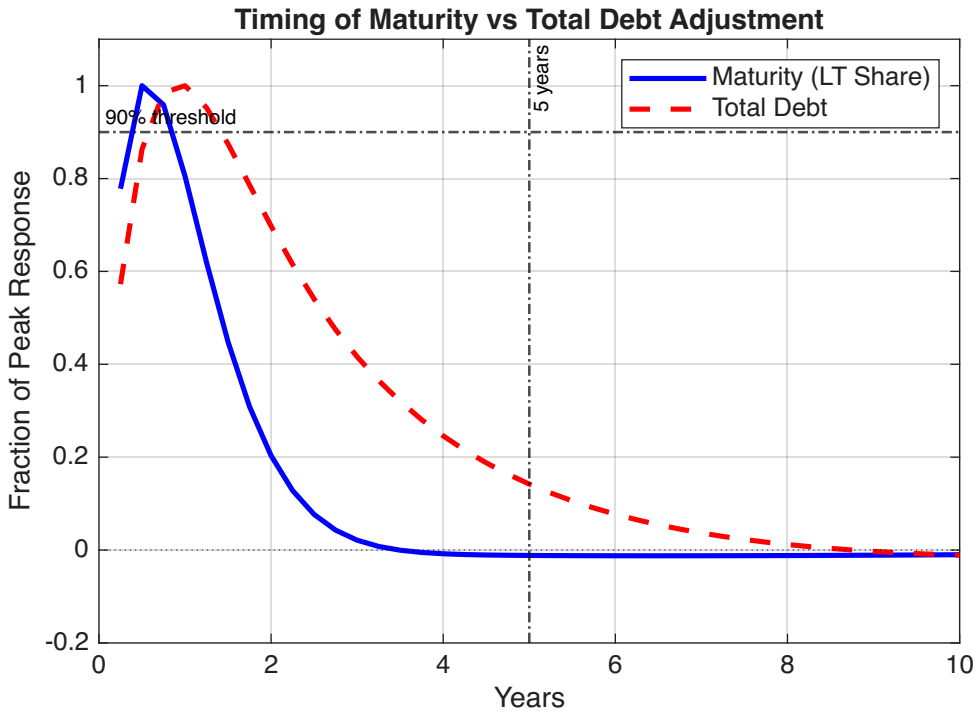

Figure 7: Timing of maturity versus total debt adjustment. Responses are normalised to peak values. Maturity reaches 90% of its peak within 2 quarters; total debt reaches 90% in 3 quarters. The vertical line marks the 5-year horizon at which empirical leverage effects become significant.

Why maturity adjusts quickly. Maturity rebalancing is a *portfolio reallocation* that does not require changing total financing. Firms shift refinancing toward longer maturities or issue bonds to repay commercial paper—adjustments involving transaction costs but not fundamental changes to investment plans. The moderate persistence $\rho_m = 0.667$ captures this flexibility.

Why leverage adjusts slowly. Increasing total debt requires new investment projects or relaxation of financing constraints, both of which operate slowly. Under NIRP, the reserve-carry drag attenuates bank net worth accumulation, further slowing the expansion of credit supply. The high persistence, $\rho_b = 0.889$, and the attenuated credit supply channel capture these frictions.

Role of policy persistence. The empirical finding that leverage rises significantly only after five years of NIRP has a natural interpretation in the model: a single transitory shock produces temporary yield compression, encouraging maturity rebalancing but dissipating before leverage fully adjusts. The model's IRFs show total debt reaching 90% of peak within 3 quarters for a *single* shock; the 5-year empirical horizon reflects the cumulative effect of *persistent* negative rates over 2014–2019, which the model rationalises through high leverage adjustment costs ($\phi_b = 8.0$).

In summary, the model provides a coherent framework rationalising the empirical evidence. Negative rates compress long-term yields through the signalling channel, inducing immediate maturity rebalancing, while total leverage responds gradually as credit supply constraints ease and firms undertake costly balance sheet adjustments.

6.13 Linearity Caveat and the NIRP Floor

The model implements the NIRP floor using Dynare's OccBin toolkit (Guerrieri and Iacoviello, 2015):

$$R_t^S = \max\{\underline{R}, R_t^{Pol}\}, \tag{35}$$

where $\underline{R} = 0.99749$ (approximately -1.0% annualised), calibrated to the reversal rate estimate in Brunnermeier and Koby (2018). With a steady-state reserve rate of $R^{ss} = 0.999$ (-0.4% annualised), the gap to the floor is approximately 15 basis points quarterly. A 25bp shock pushes the Taylor rule prescription below the floor, triggering binding for one quarter. Figure 5 illustrates this mechanism.

6.14 Counterfactual Analysis: Gradual versus Single Rate Cuts

A natural policy question is whether the pace of rate cuts matters for transmission. The ECB implemented NIRP through a series of gradual reductions—from -0.10% in June 2014 to -0.50% by September 2019—rather than a single large cut. We use the model's linear structure

to conduct a counterfactual analysis comparing a single 25 basis point cut at $t = 1$ (our baseline) with three gradual cuts of 8.3 basis points each at $t = 1, 2, 3$ (the counterfactual). Figure 8 presents the results across nine key variables. The main findings are:

Similar cumulative output effects. Output peaks at 0.119% in quarter 2 under the single cut versus 0.110% in quarter 3 under gradual cuts. Cumulative output over 40 quarters differs by less than 1%, suggesting that the timing of cuts matters little for total stimulus.

Different binding behaviour. The most striking difference is whether the NIRP floor binds. Under the single 25bp cut, the floor binds for one quarter, activating the forward guidance channel and causing net worth to rise on impact (+0.0031%). Under gradual cuts of 8.3bp each, no single cut is large enough to trigger the floor (since the gap is 15bp), so the floor never binds. Without the forward guidance effect, net worth dynamics follow a more conventional pattern. This demonstrates that the *size* of rate cuts matters for NIRP transmission, not just their cumulative magnitude.

Smoother interest rate adjustment. Both the real short rate and real long yield adjust more gradually under staggered cuts. The real short rate falls by 0.33% on impact under the single cut versus 0.12% under gradual cuts, and the real long yield falls by 0.18% versus 0.06%. This smoother adjustment reduces financial market volatility and gives institutions time to reposition portfolios.

Preserved debt structure dynamics. The "borrow long, dump short" pattern is preserved under both scenarios. The long-term debt share rises by 0.12 percentage points on impact under the single cut versus 0.04 pp under gradual cuts, but both converge to similar cumulative levels (approximately 0.77 pp-quarters over 40 quarters).

These counterfactual results reveal a trade-off in NIRP implementation. A single large cut that triggers the floor activates the forward guidance channel, generating immediate net worth gains and stronger front-loaded stimulus. Gradual cuts avoid floor binding entirely, producing smoother adjustment without amplifying forward guidance. The choice depends on policymakers' objectives: if immediate stimulus is paramount, larger cuts that trigger the floor may be preferable; if smooth adjustment and predictability are valued, gradual cuts may be preferred.

6.15 Robustness: Sensitivity to Adjustment Cost Parameters

The differential timing between maturity and leverage adjustment is driven by the calibrated adjustment cost parameters $\phi_m = 2.0$ and $\phi_b = 8.0$, implying a 4:1 ratio. To assess robustness,

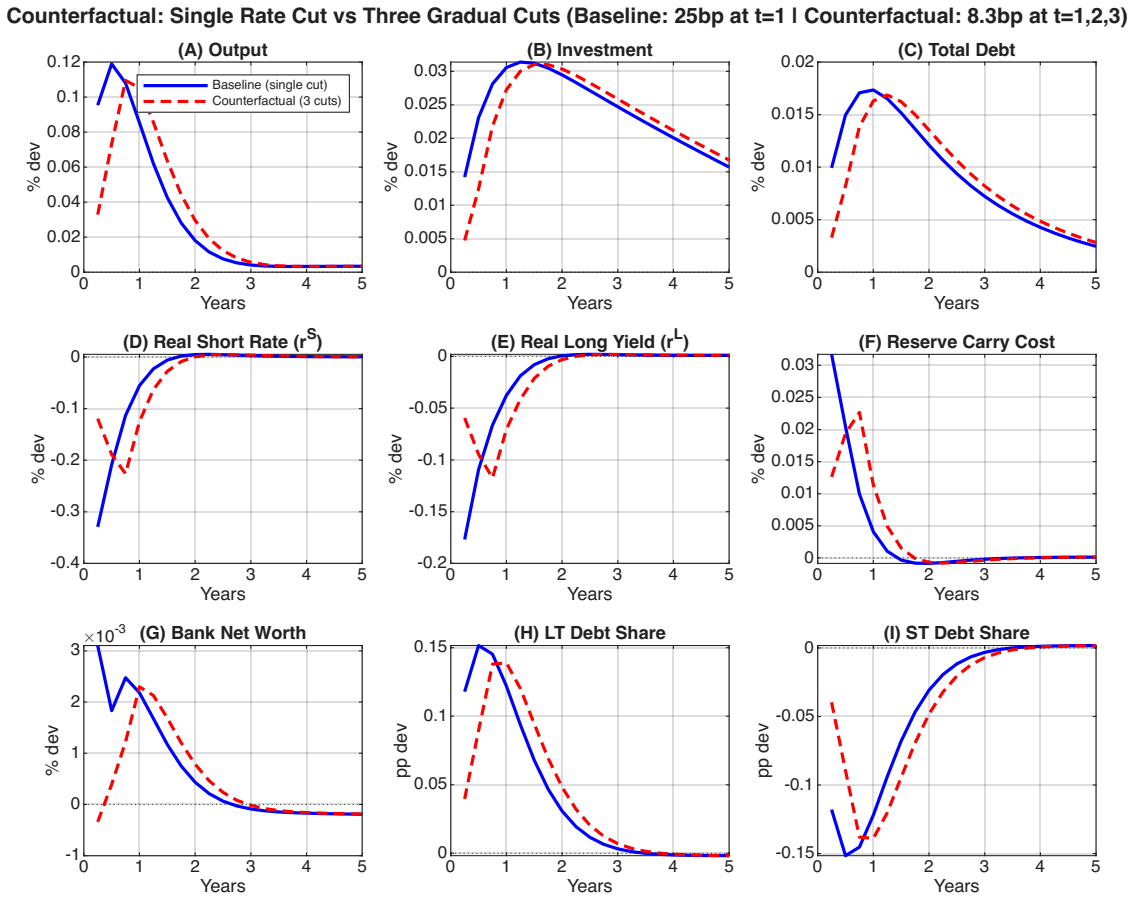

Figure 8: Counterfactual analysis: single rate cut versus three gradual cuts. Baseline (solid blue): full 25bp cut at $t = 1$, which triggers the NIRP floor and generates positive net worth on impact (+0.0031%). Counterfactual (dashed red): three 8.3bp cuts at $t = 1, 2, 3$, none of which triggers the floor, resulting in negative net worth on impact (−0.0003%). Row 1: Real economy variables. Row 2: Interest rates. Row 3: Financial and debt structure variables. Both scenarios achieve similar cumulative output effects (differing by less than 1%).

we vary $\phi_m \in \{1.0, 2.0, 4.0\}$ and $\phi_b \in \{4.0, 8.0, 16.0\}$ and report the time to 90 percent of peak response (t_{90}) for both maturity and total debt.

Table 17: Robustness Analysis: Time to 90% of Peak Response (Quarters)

		ϕ_b		
		4.0	8.0	16.0
ϕ_m	1.0	$t^m_{90} = 1$, $t^b_{90} = 2$	$t^m_{90} = 1$, $t^b_{90} = 3$	$t^m_{90} = 1$, $t^b_{90} = 3$
	2.0	$t^m_{90} = 2$, $t^b_{90} = 2$	$t^m_{90} = 2$, $t^b_{90} = 3$	$t^m_{90} = 2$, $t^b_{90} = 3$
	4.0	$t^m_{90} = 2$, $t^b_{90} = 2$	$t^m_{90} = 2$, $t^b_{90} = 3$	$t^m_{90} = 2$, $t^b_{90} = 3$

Notes: Each cell reports the quarter in which maturity (t^m_{90}) and total debt (t^b_{90}) reach 90 percent of their peak response following a 25bp NIRP shock. The baseline calibration ($\phi_m = 2.0$, $\phi_b = 8.0$) is highlighted. The key finding—that maturity adjusts faster than or equal to leverage ($t^m_{90} \leq t^b_{90}$)—is robust across all parameter combinations.

The key finding — that maturity adjusts faster than leverage — is robust across all parameter combinations. In every case, $t^m_{90} \leq t^b_{90}$. The qualitative ordering is preserved even when ϕ_m is raised to 4.0 (making maturity adjustment costly) or when ϕ_b is lowered to 4.0 (making leverage adjustment easier).

However, the quantitative timing differential does depend on the ratio ϕ_b/ϕ_m, as expected from the structural derivation. When $\phi_b/\phi_m = 2$ (e.g., $\phi_m = 2$, $\phi_b = 4$), both adjust in similar time frames ($t^m_{90} = t^b_{90} = 2$). When $\phi_b/\phi_m \geq 4$, the timing differential emerges with maturity adjusting faster. Our baseline ratio of 4:1 produces a differential of one quarter ($t^m_{90} = 2$ vs $t^b_{90} = 3$), consistent with the empirical evidence that maturity responds earlier while leverage responds over years.

This robustness exercise confirms that the model's central mechanism — differential adjustment speeds arising from heterogeneous adjustment costs — is not knife-edge dependent on particular parameter values. The qualitative prediction of faster maturity adjustment is preserved across economically plausible calibrations.

7 Conclusion and Policy Implications

This study examines the impact of the European Central Bank's negative interest rate policy introduced in 2014 on corporate financing structures. We hypothesise that firms operating under NIRP restructure their debt maturities by shifting from short-term to long-term borrowing to lock in historically low interest rates over extended horizons. To test this hypothesis, we employ a difference-in-differences framework in which Eurozone firms subject to ECB policy constitute the treatment group and firms in non-NIRP countries serve as controls.

We implement multiple estimation strategies to ensure robustness. The traditional two-way fixed effects estimator, appropriate for our two-group, two-period setting, is complemented by the heterogeneous treatment effects method of Callaway and Sant'Anna (2021) and the matching-based approach of Imai et al. (2023), which strengthens the plausibility of the parallel trends assumption. Across all specifications, we find consistent evidence that firms reduce short-term debt while increasing long-term debt in response to NIRP — a pattern we characterise as "borrow long, dump short." Subsample analysis reveals that this maturity rebalancing is most pronounced among smaller, older, more leveraged, and less liquid firms. Crucially, however, the shift in debt maturity does not translate into an immediate increase in total firm leverage. We find that leverage expands significantly only when negative rates are sustained for approximately 5 years.

To interpret these empirical findings, we develop a medium-scale DSGE model calibrated to Eurozone data that incorporates financial intermediaries, endogenous corporate debt maturity choice, and the institutional features of NIRP — notably, the zero lower bound on retail deposit rates that creates a wedge between policy rates and deposit rates. The model identifies two principal transmission channels.

First, the signalling channel operates through term structure compression. When the central bank cuts the reserve rate into negative territory, forward-looking bond pricing implies a sharp decline in real long-term yields. This compression reduces the relative cost of long-maturity financing, inducing firms to substitute toward long-term debt. The model predicts that maturity rebalancing occurs rapidly, consistent with our empirical evidence of immediate maturity restructuring following NIRP adoption.

Second, the reserve-carry channel governs total leverage dynamics. The binding deposit floor imposes a margin squeeze on banks, but under an accommodative policy stance, capital gains on long-term bond portfolios dominate, so that bank net worth rises on impact and supports credit supply. This attenuation relative to conventional easing explains why total debt adjusts gradually, consistent with our empirical finding that leverage rises significantly only after prolonged exposure to negative rates.

The distinction between rapid maturity adjustment and delayed leverage expansion has a natural economic interpretation. Maturity rebalancing is fundamentally a portfolio reallocation decision: firms can shift refinancing toward longer maturities without changing their total borrowing volume. Such adjustments involve transaction costs but do not require new investment projects or relaxation of financing constraints. Leverage expansion, by contrast, requires either increased investment demand or easing of credit supply conditions—both of which operate on longer time horizons.

From a policy perspective, our results carry important implications for the design and communication of unconventional monetary policy. Negative interest rates are more likely to stimulate

aggregate corporate borrowing when implemented persistently and accompanied by clear forward guidance. A transitory NIRP shock may successfully induce maturity rebalancing — as firms opportunistically lock in low long-term rates — but will fail to generate meaningful leverage expansion if market participants perceive the policy as temporary.

Our counterfactual analysis further suggests that the pace of rate cuts matters: larger cuts that trigger the NIRP floor activate the forward guidance channel and amplify the stimulus, while gradual cuts produce smoother adjustment at the cost of forgoing this amplification. The ECB's incremental approach from -0.10% to -0.50%, combined with explicit forward guidance, appears well-suited to supporting credit growth, though with substantial lags.

Finally, our results highlight the importance of considering debt structure — not merely debt levels — when evaluating monetary policy transmission. The rapid maturity rebalancing we document has implications for corporate interest rate exposure, refinancing risk, and the sensitivity of firm cash flows to future rate changes. As central banks contemplate the normalisation of policy rates, understanding how the corporate sector has repositioned its debt maturity during the low-rate era will be essential for assessing the financial stability implications of monetary tightening

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